Enclustra User Schematic

Andromeda PE5 Revision 1.1

Disclaimer

User schematics do not include proprietary Enclustra design elements, such as power supply circuits.
These circuits use optimized designs which are extremely compact and remain proprietary to Enclustra.
The user manual provides all necessary information to use the module and its interfaces.

User schematics are provided after purchase of Enclustra hardware, and may also be provided in certain cases before purchase, to assist in product evaluation.
No part of the schematics may be copied or modified without written permission from Enclustra.

Full schematics, including a full bill-of-materials, may be available through the purchase of a hardware licence for the product in question.
Please contact Enclustra Sales for more information.

Note: DNE = Do Not Equip (parts not equipped by default)

© 2025 by Enclustra FPGA Solutions

Andromeda PE5 Base Board

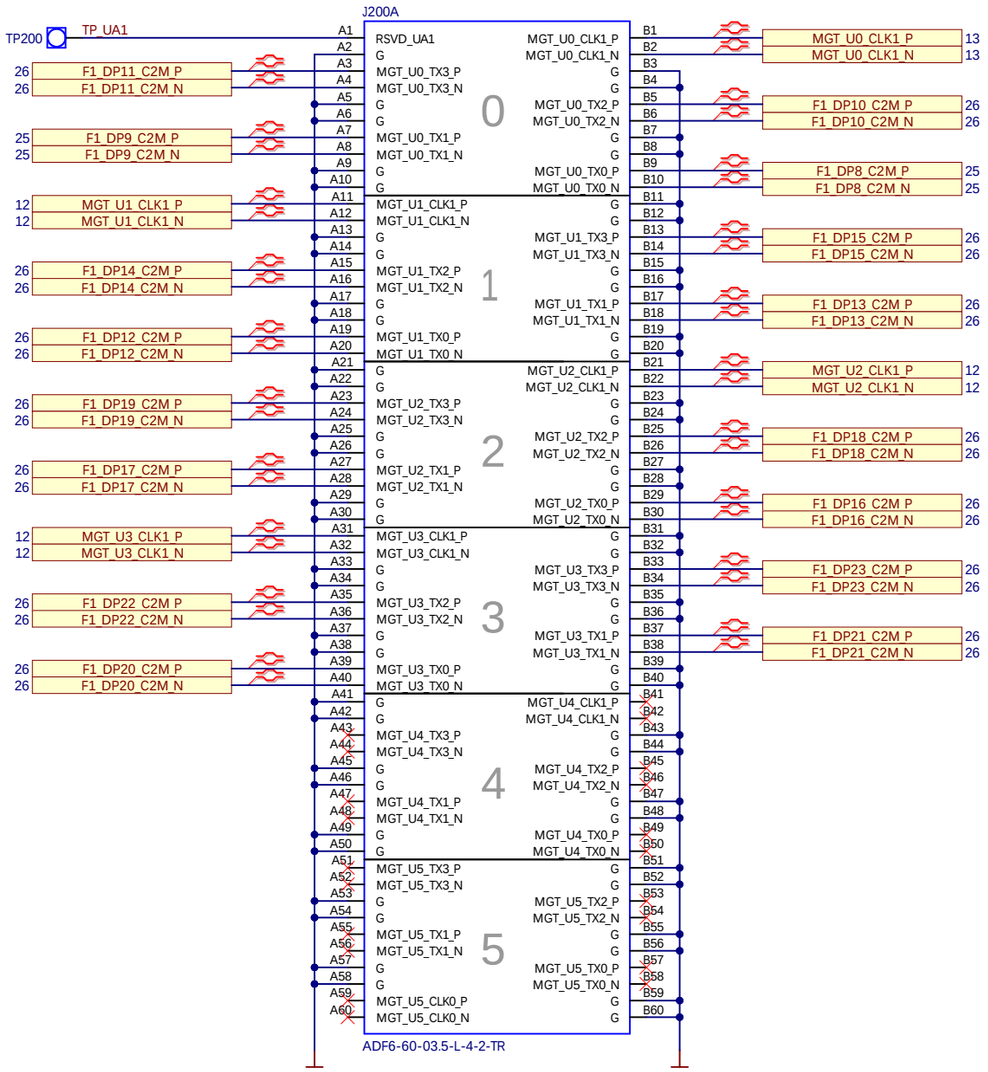
Revision 1.1

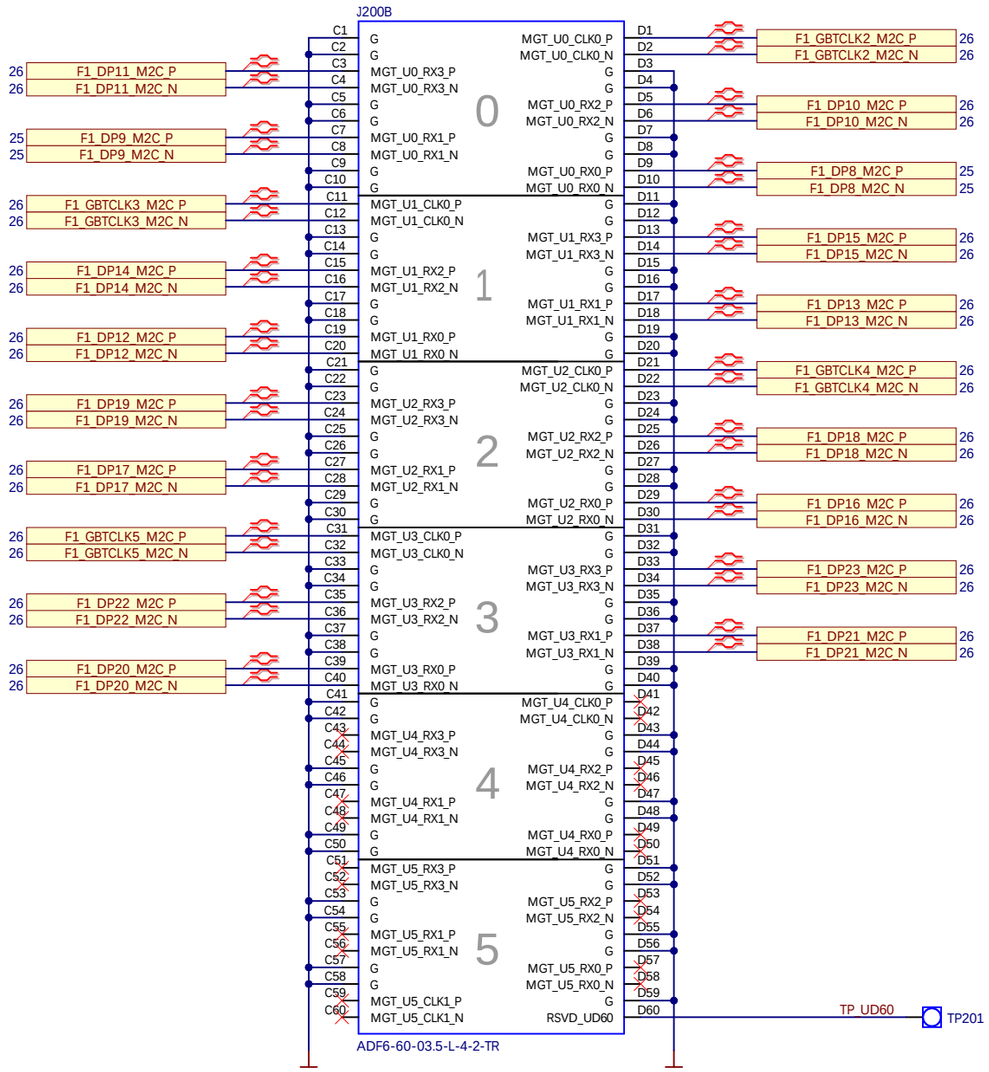
Sheet 01:	Cover	Sheet 19:	HDMI Input
Sheet 02:	Connector U	Sheet 20:	SDI Outputs
Sheet 03:	Connector V	Sheet 21:	SDI Inputs | Digital IO
Sheet 04:	Connector W	Sheet 22:	PCIe x8 Connector
Sheet 05:	Connector X	Sheet 23:	FMC 0 Connector Part 1
Sheet 06:	Connector Y	Sheet 24:	FMC 0 Connector Part 2
Sheet 07:	Connector Z	Sheet 25:	FMC 1 Connector Part 1
Sheet 08:	USB 2.0 Device Controller	Sheet 26:	FMC 1 Connector Part 2
Sheet 09:	I2C | User EEPROM	Sheet 27:	JTAG | CTRL | PMOD | Fan
Sheet 10:	USB Configuration	Sheet 28:	User Interface
Sheet 11:	USB 3.0	Sheet 29:	Power Input | 3V3_AUX
Sheet 12:	Clock Generator 0	Sheet 30:	Power 3V3 | IO_F0 | ADJ_F0 | 5V
Sheet 13:	Clock Generator 1 | Oscillator	Sheet 31:	Power 2V5 | IO_F1 | ADJ_F1 | 1V8
Sheet 14:	Clock Generator 2 | TCXO	Sheet 32:	Reset Circuit | Over-Voltage Protection
Sheet 15:	Gigabit ETH | SD-Card | M.2	Sheet 33:	Power Module
Sheet 16:	SFP28 | QSFP28	Sheet 34:	System Monitor
Sheet 17:	Display Port Interface	Sheet 35:	Mechanics
Sheet 18:	HDMI Output	Sheet 36:	Component Assembly

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P01_Cover.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 1 / 36

Connector U

MGT - MGT - MGT - MGT - NC - NC

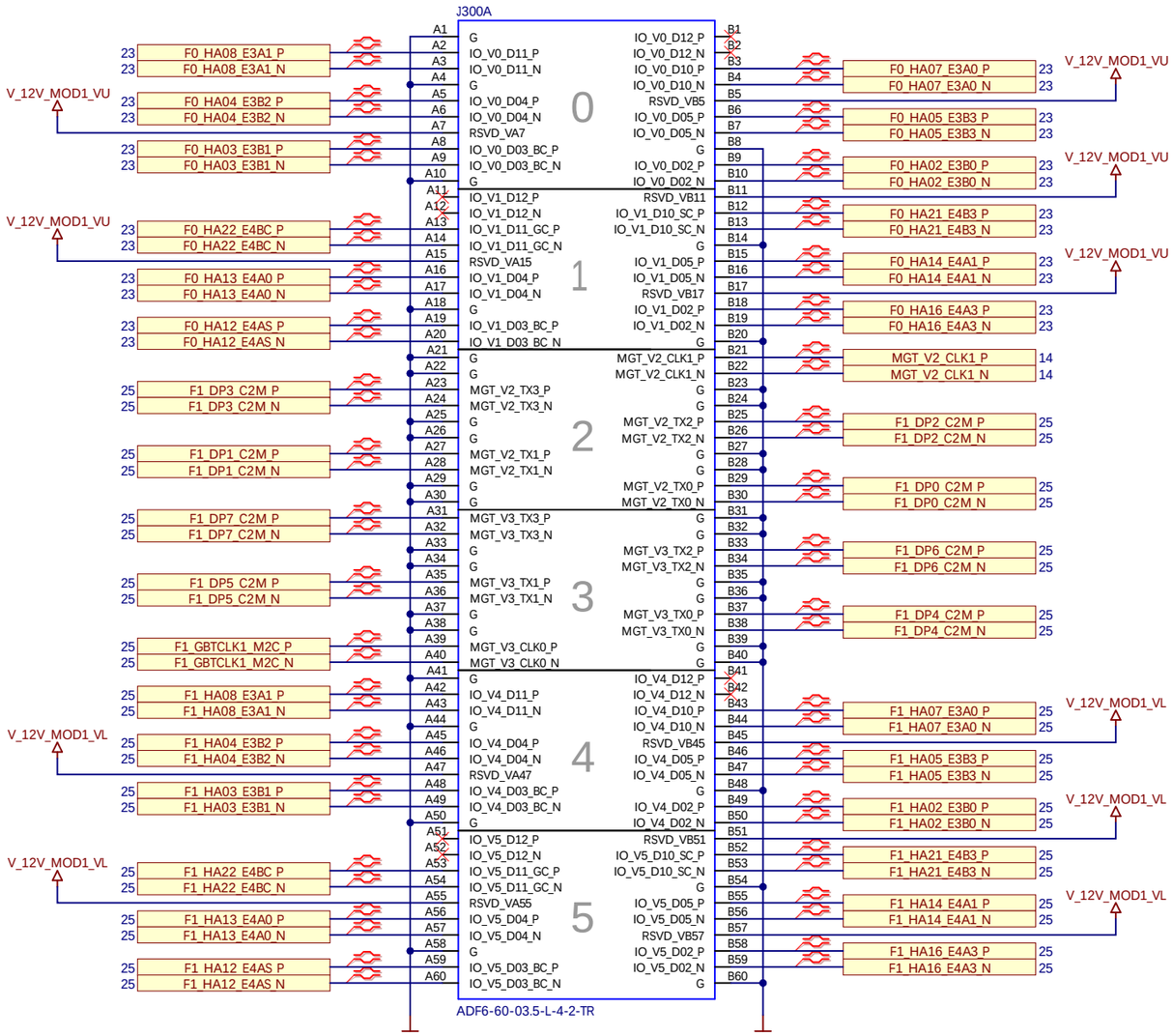


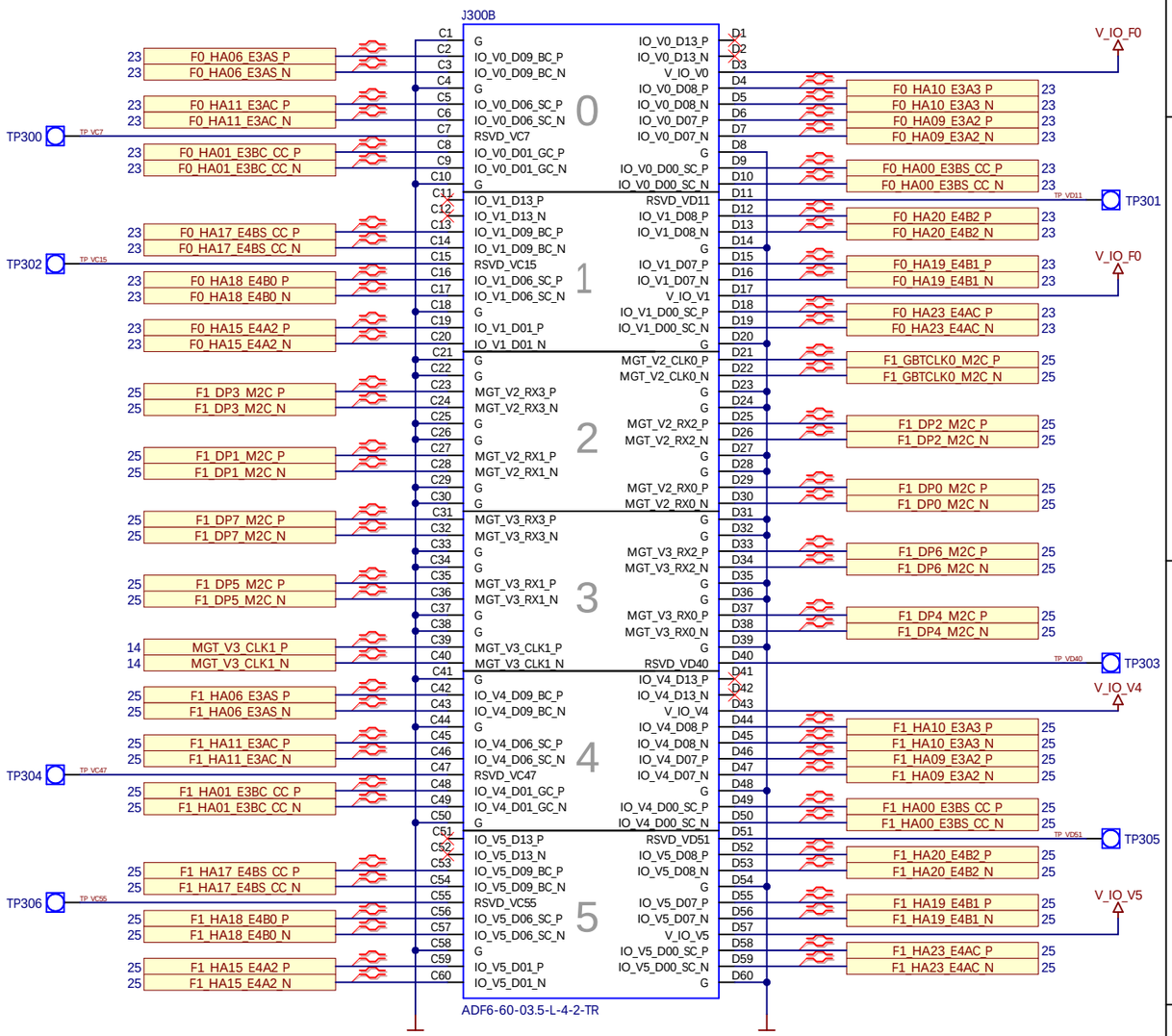


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P02_Connector_U.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 2 / 36

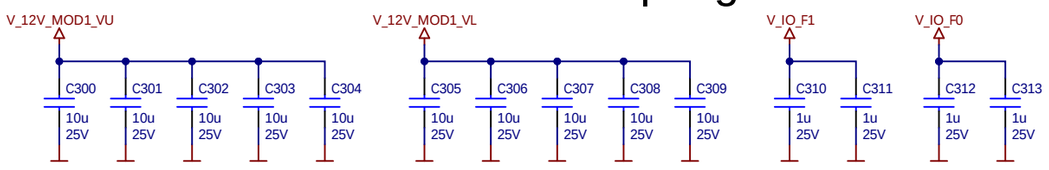
Connector V

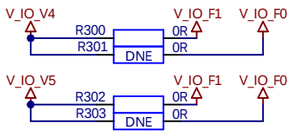
IO - IO - MGT - MGT - IO - IO





Module Decoupling

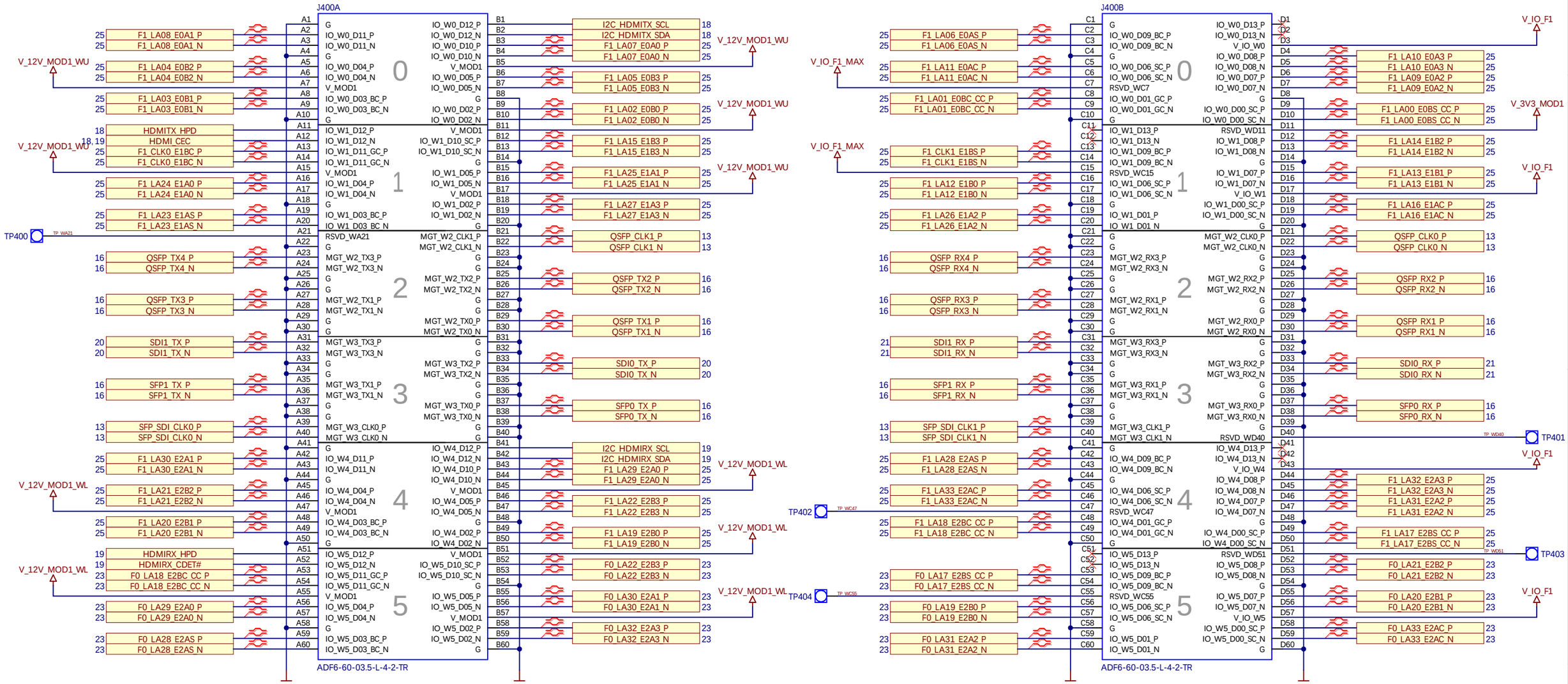




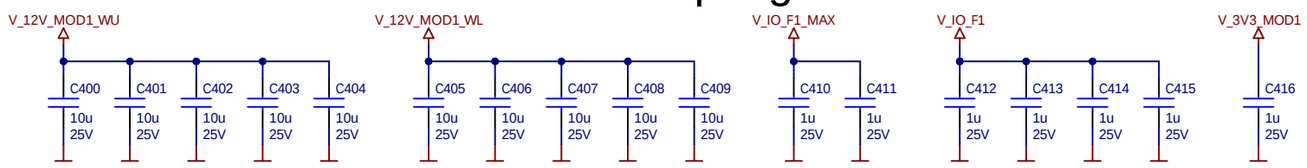
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P03_Connector_V.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 3 / 36

Connector W

IO - IO - MGT - MGT - IO - IO



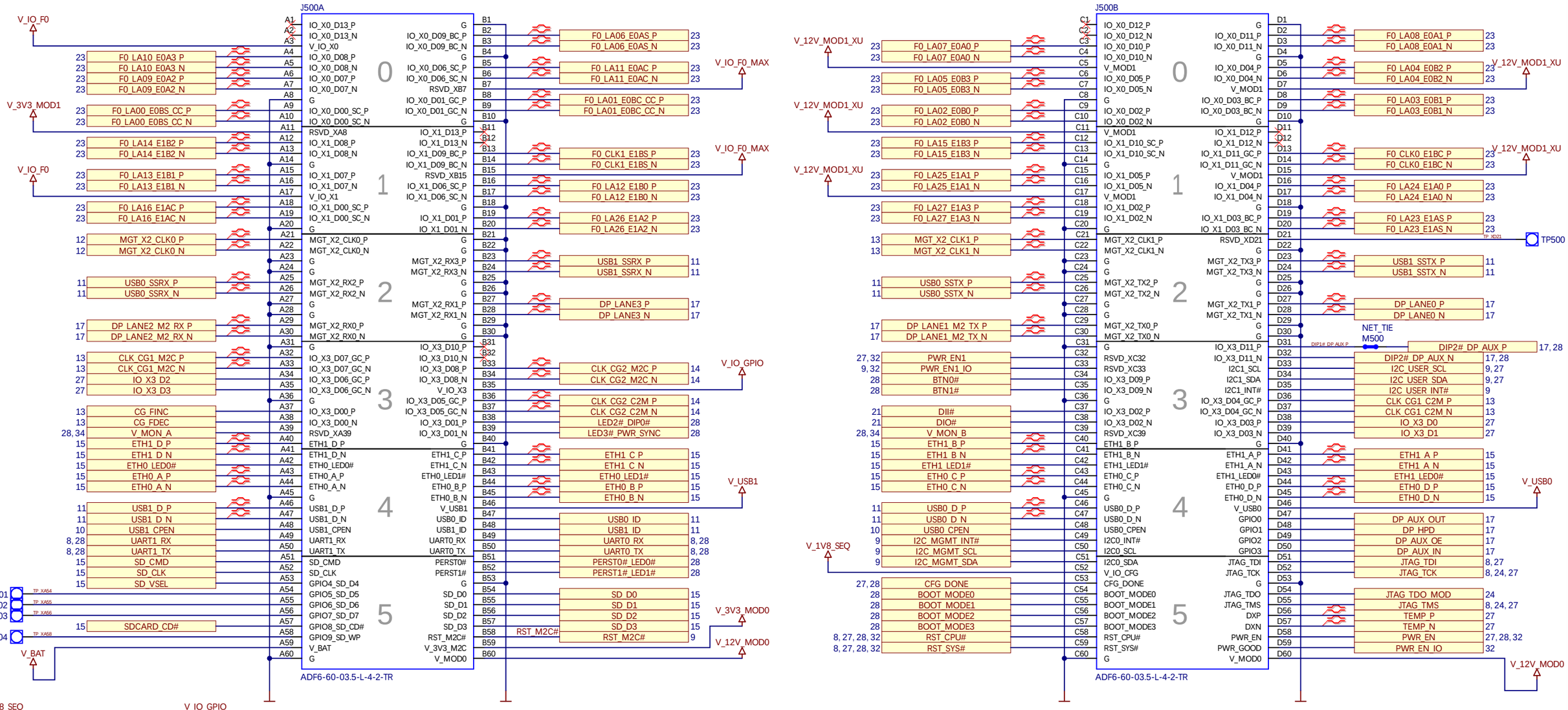
Module Decoupling



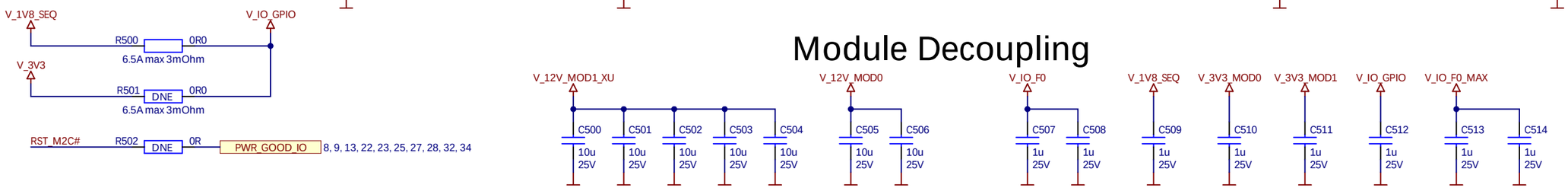
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P04_Connector_W.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 4 / 36

Connector X

IO - IO - MGT - IO - ETH/USB - Control



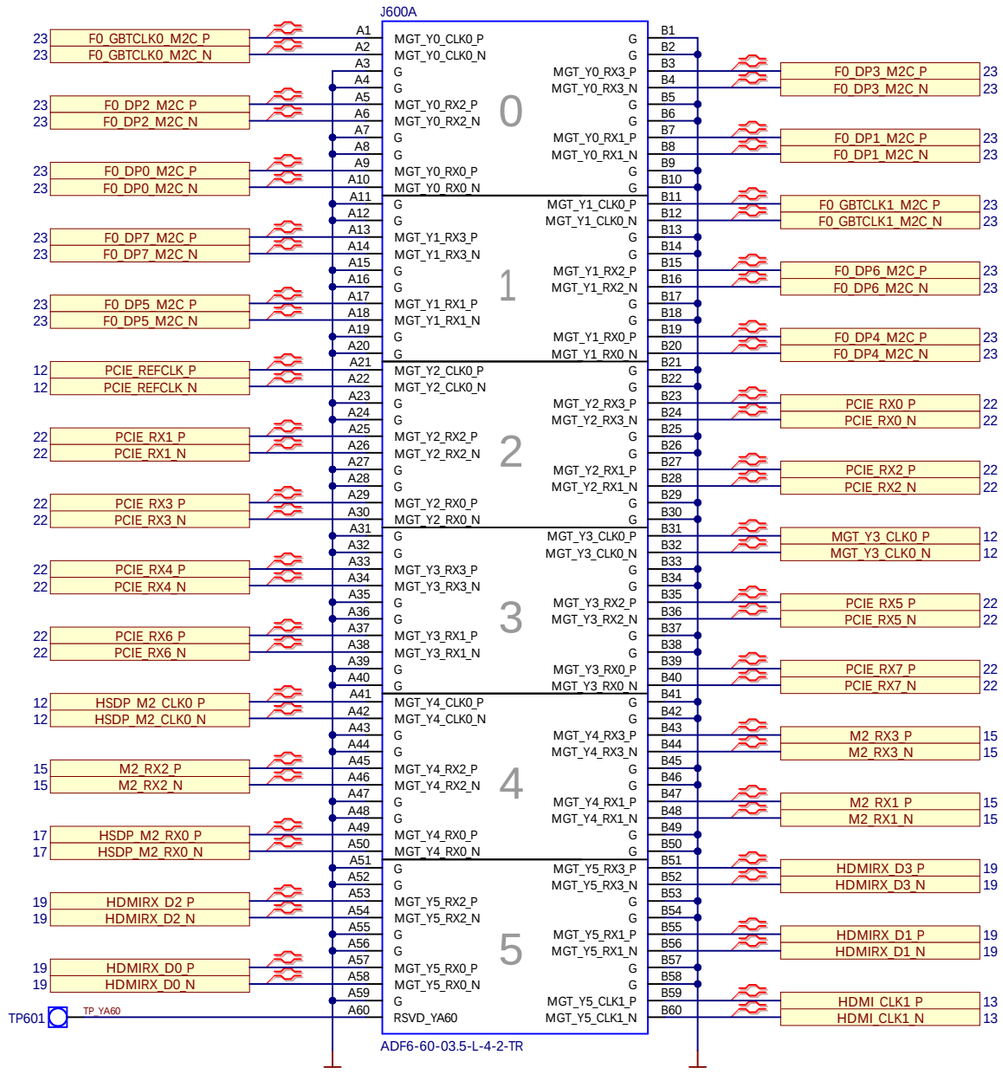
Module Decoupling

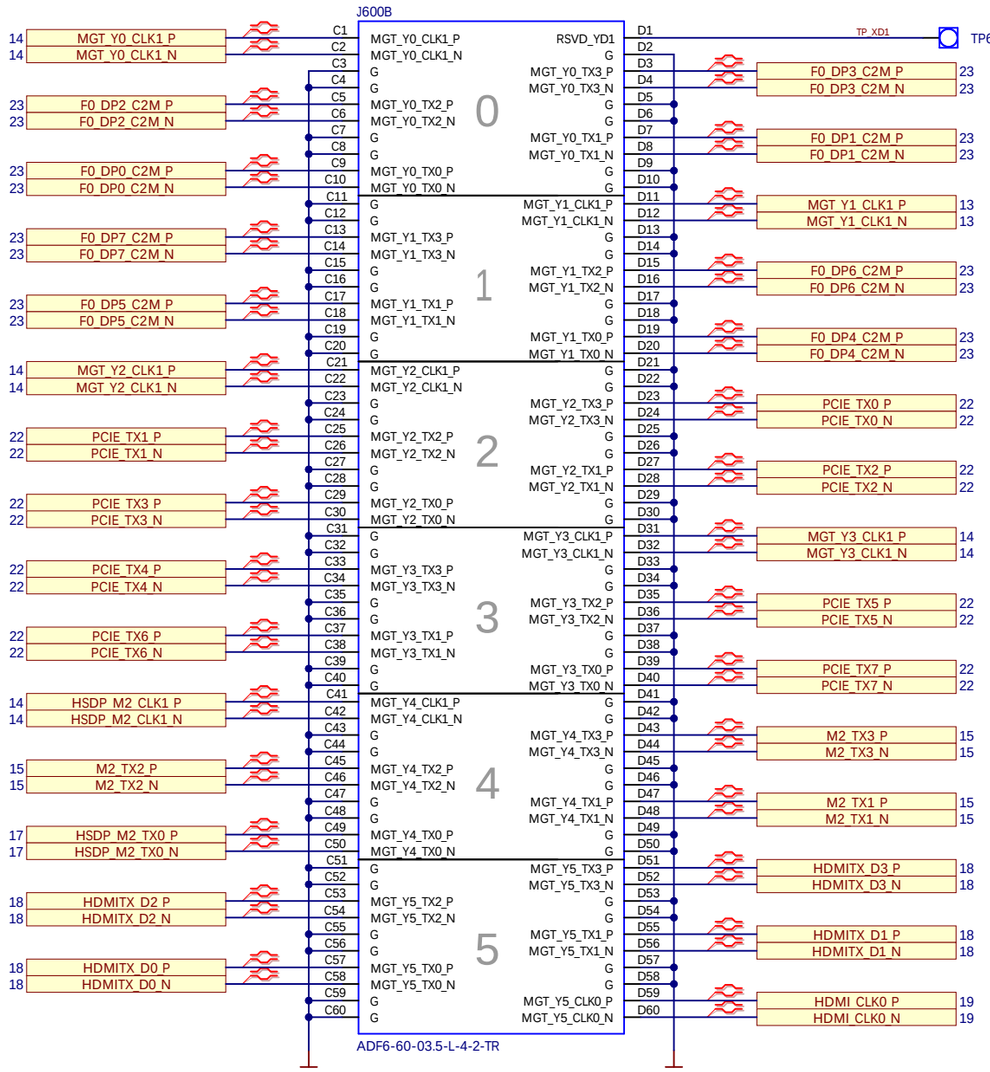


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P05_Connector_X.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 5 / 36

Connector Y

MGT - MGT - MGT - MGT - MGT - MGT

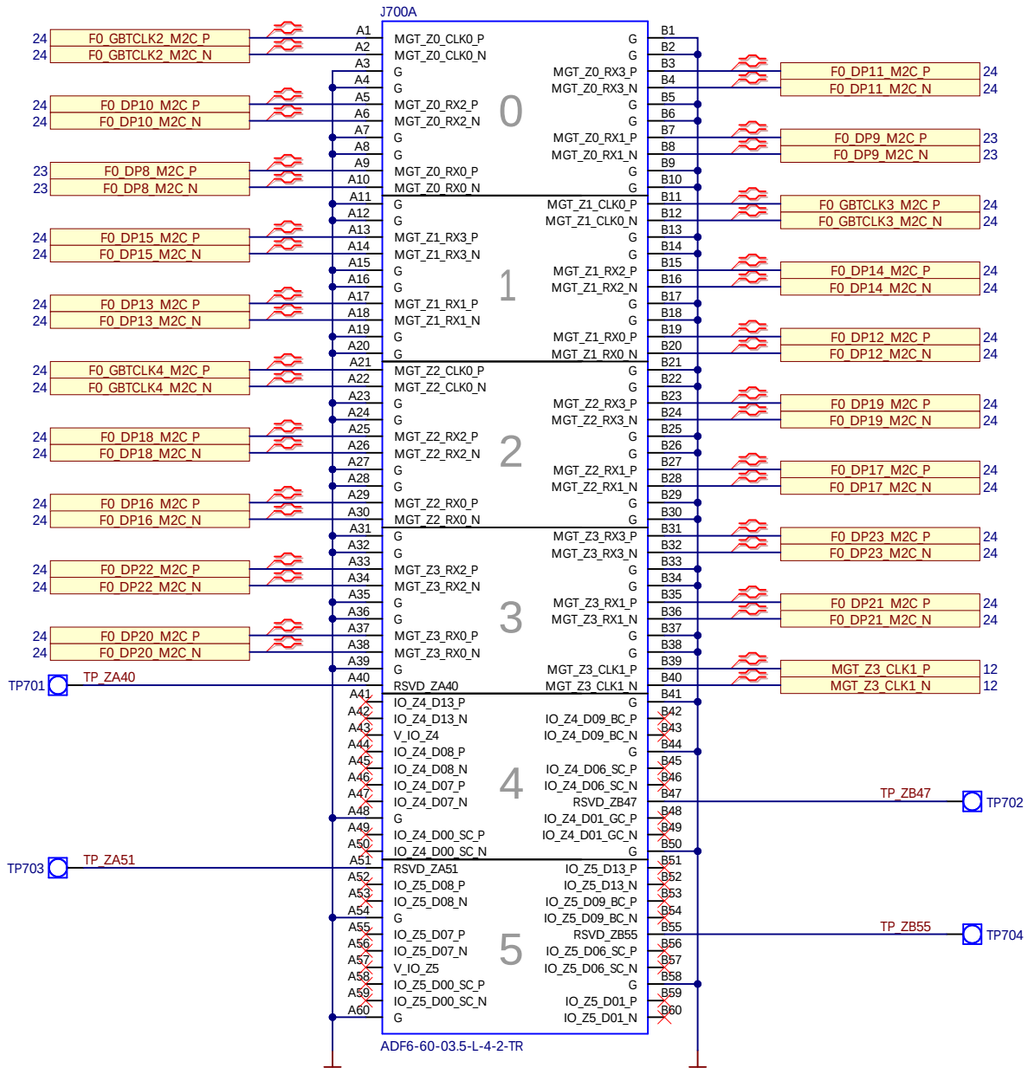


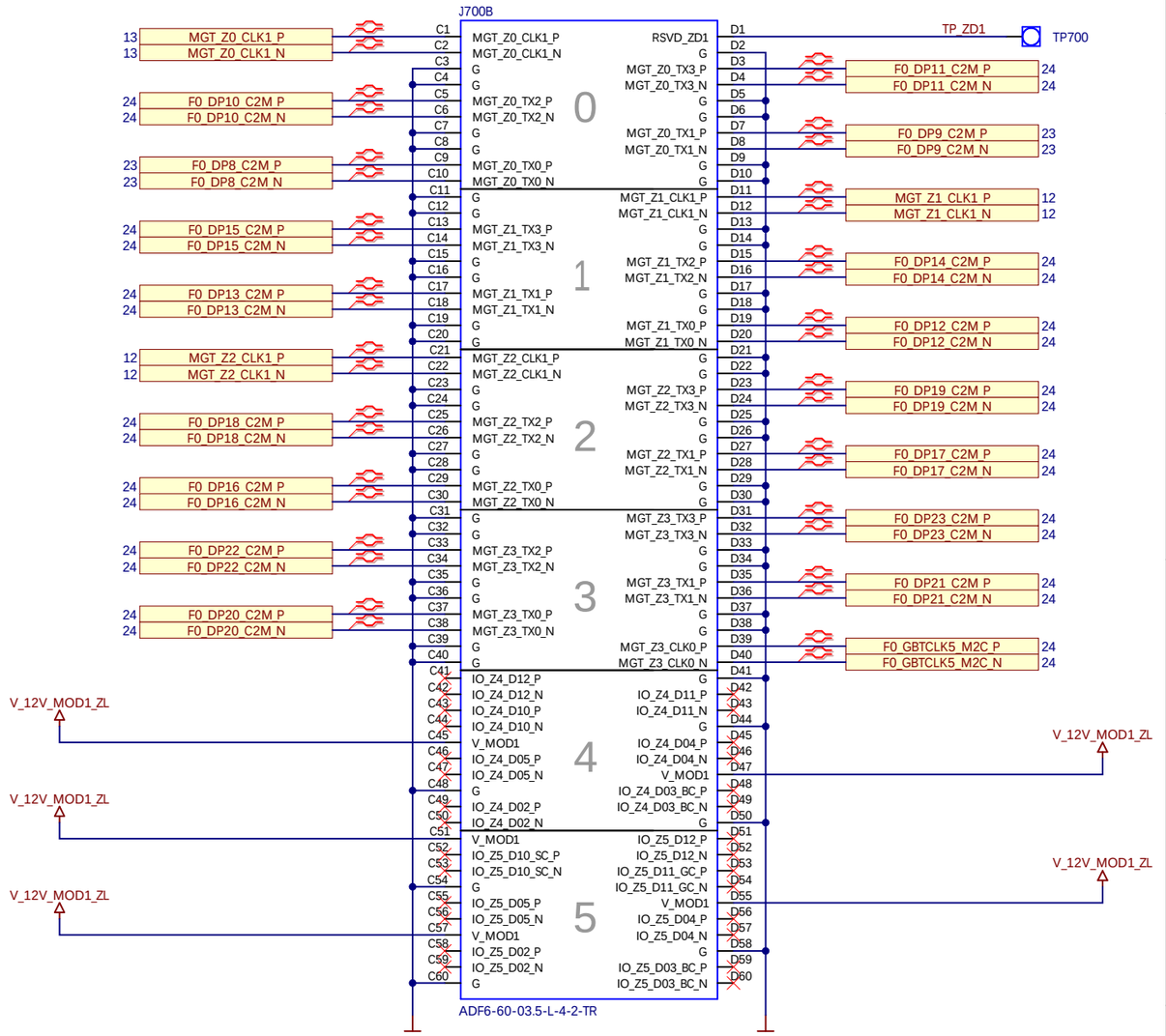


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P06_Connector_Y.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 6 / 36

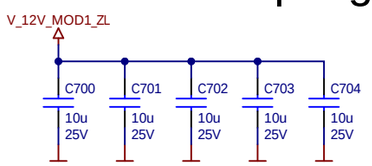
Connector Z

MGT - MGT - MGT - MGT - NC - NC



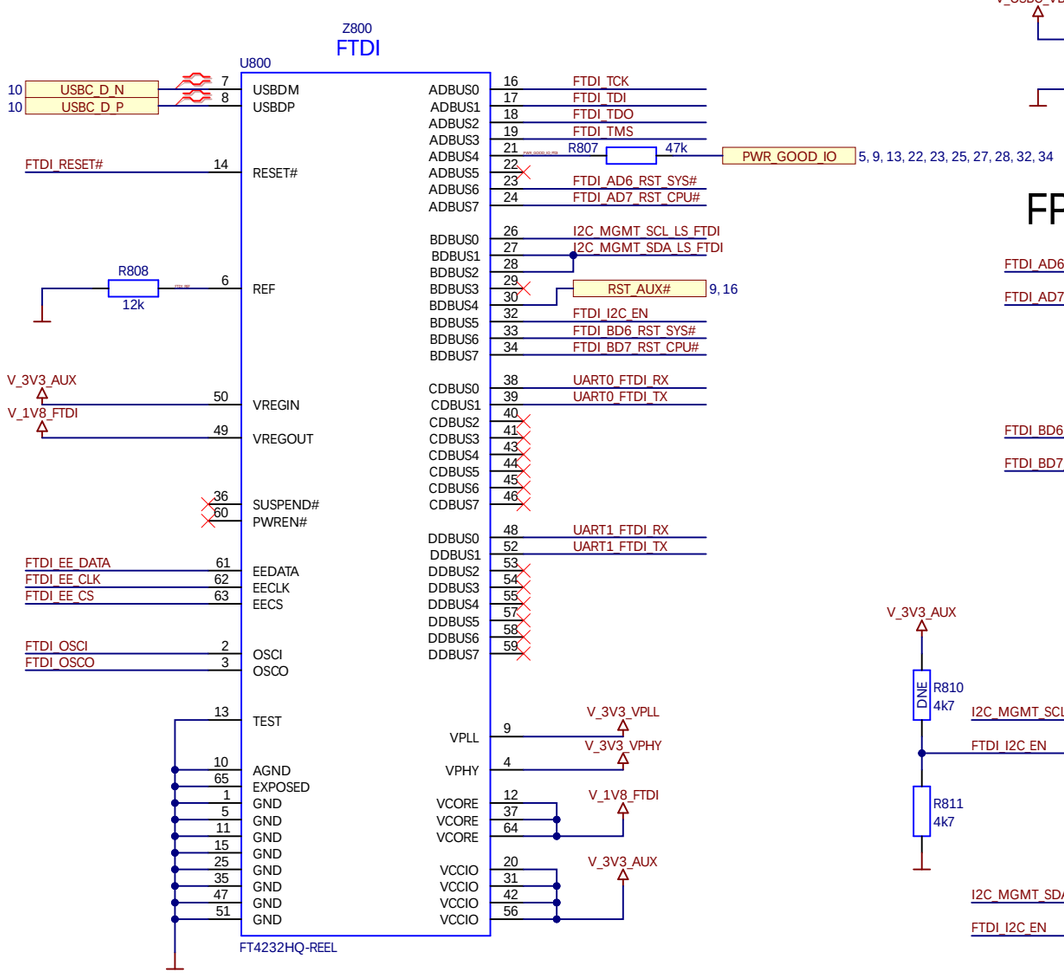


Module Decoupling

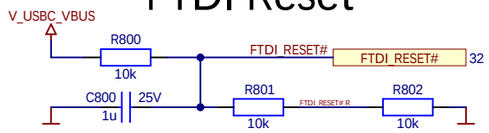


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P07_Connector_Z.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 7 / 36

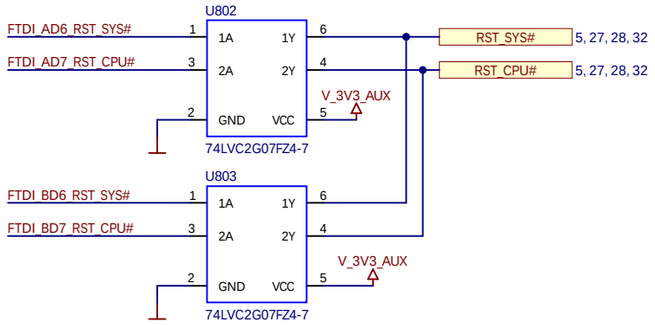
FTDI USB 2.0 Device Controller



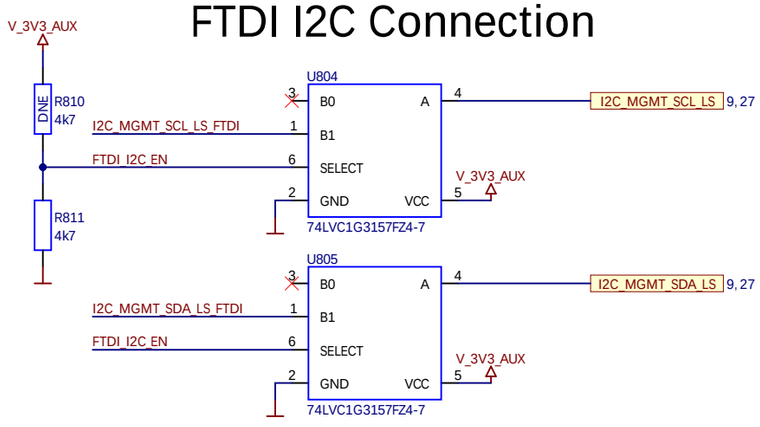
FTDI Reset



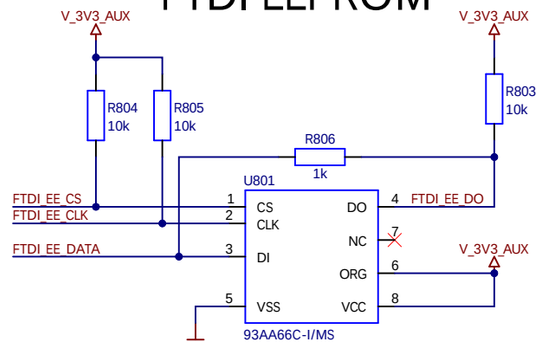
FPGA Reset Generation



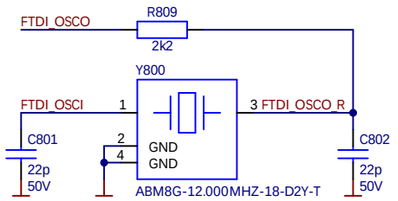
FTDI I2C Connection



FTDI EEPROM



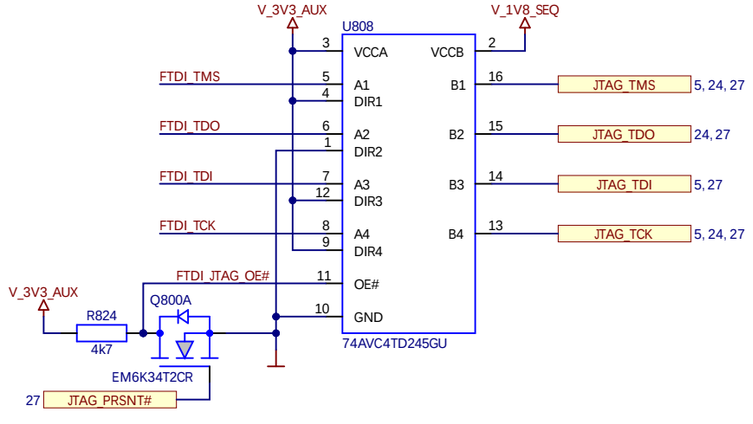
FTDI Clock



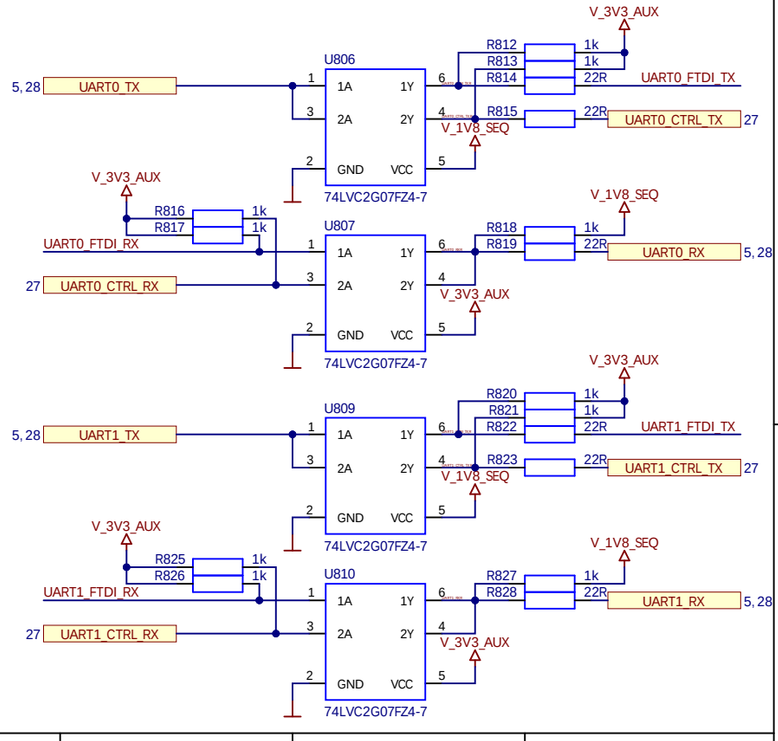
FTDI Power Filters & Decoupling

not included in user schematic

FTDI JTAG Level Shifters

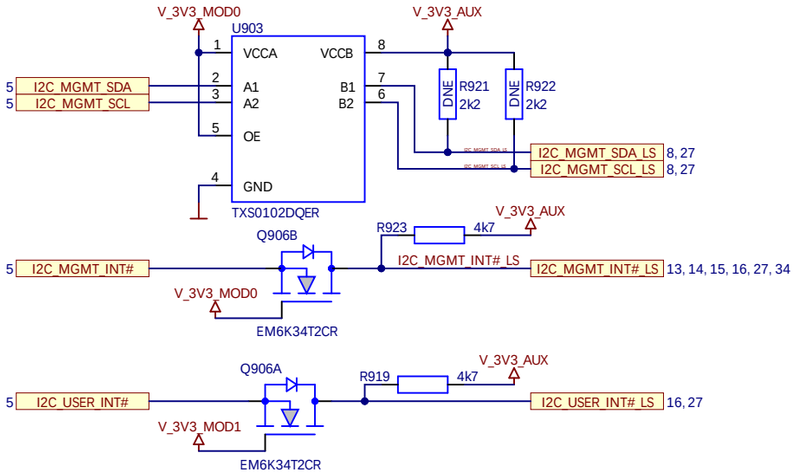


FTDI UART Level Shifters

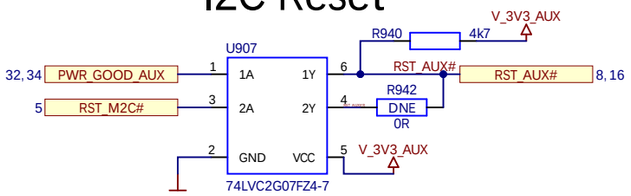


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P08_USB_Controller.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 8 / 36

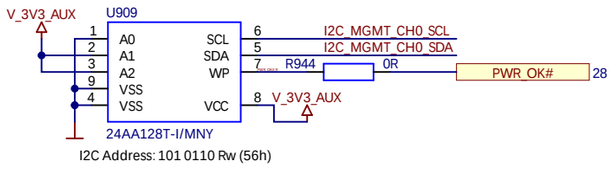
I2C Level Shifters



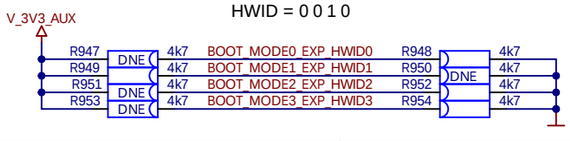
I2C Reset



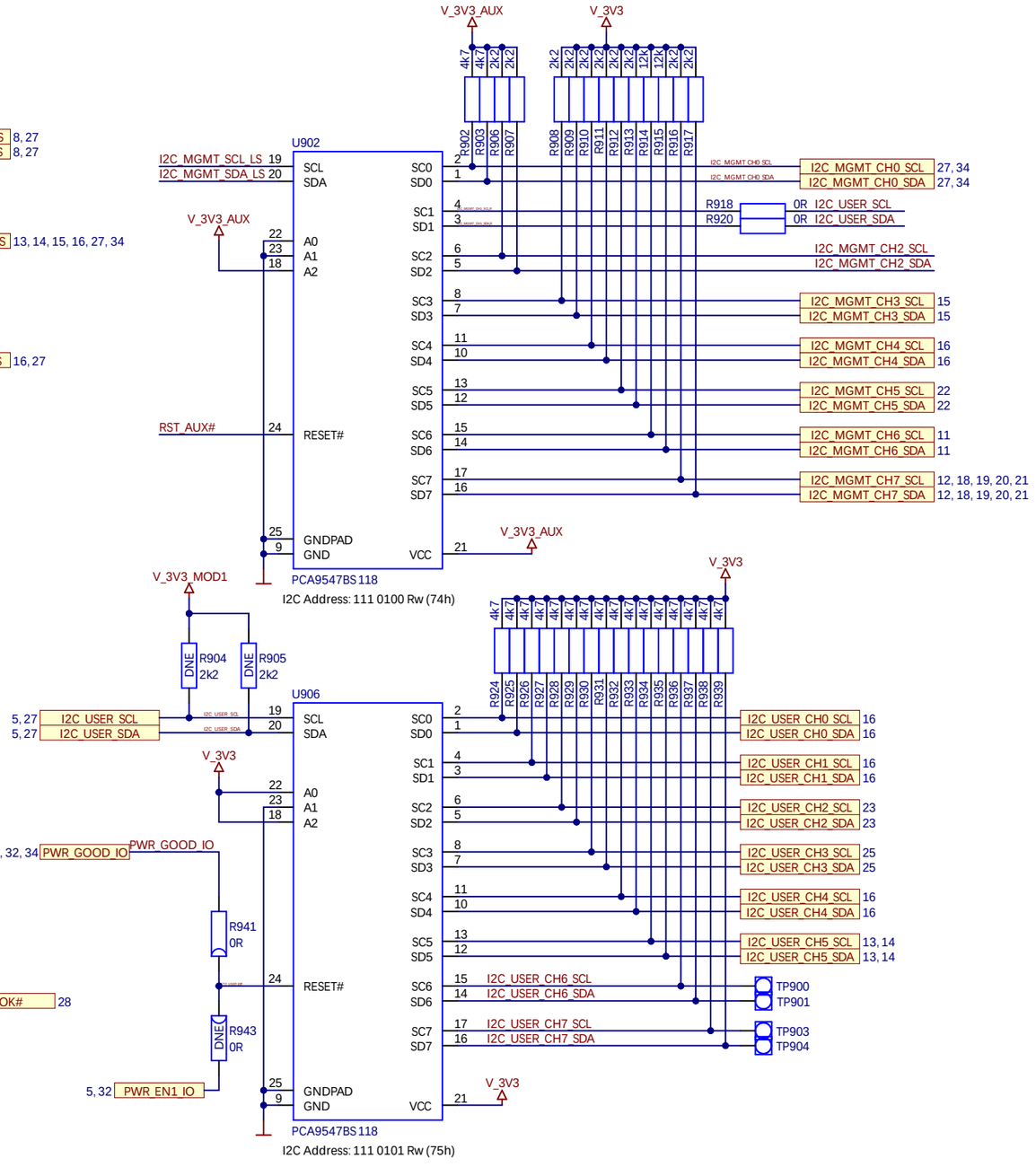
User EEPROM



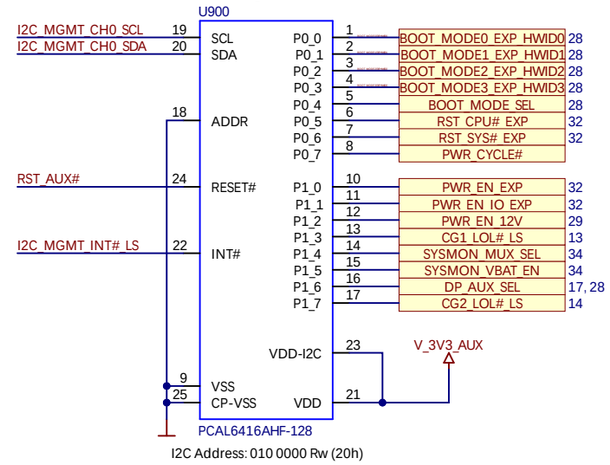
Hardware Revision



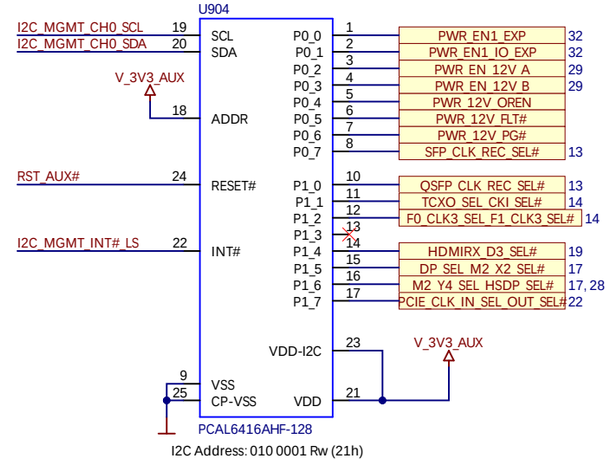
I2C Multiplexers



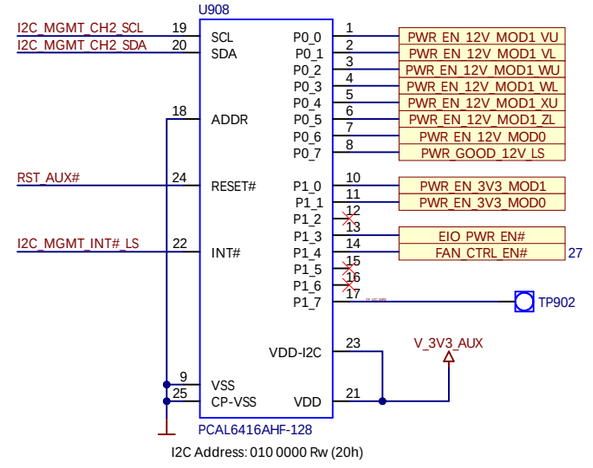
I2C IO Expander 0



I2C IO Expander 1

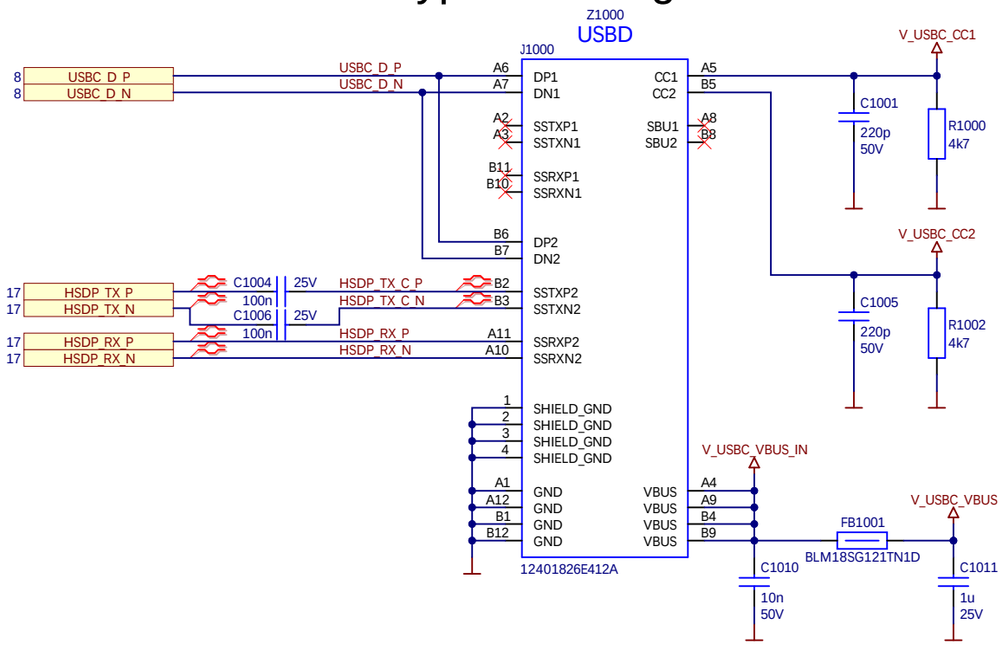


I2C IO Expander 2

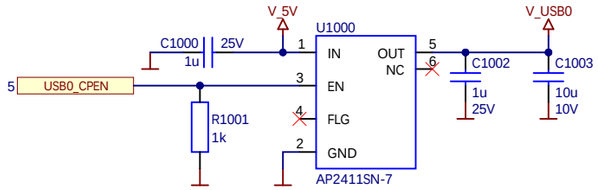


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P09_I2C_Multiplexers.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 9 / 36

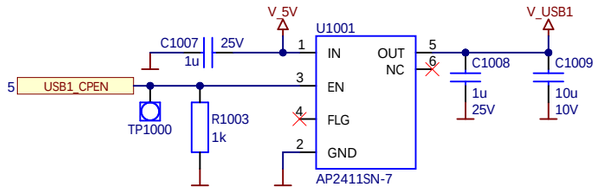
USB Type-C Config. Connector



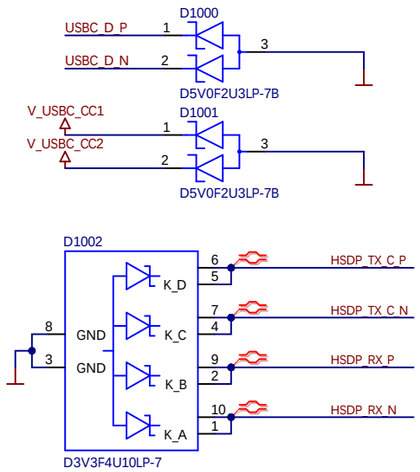
USB0 Load Switch DFP



USB1 Load Switch DFP

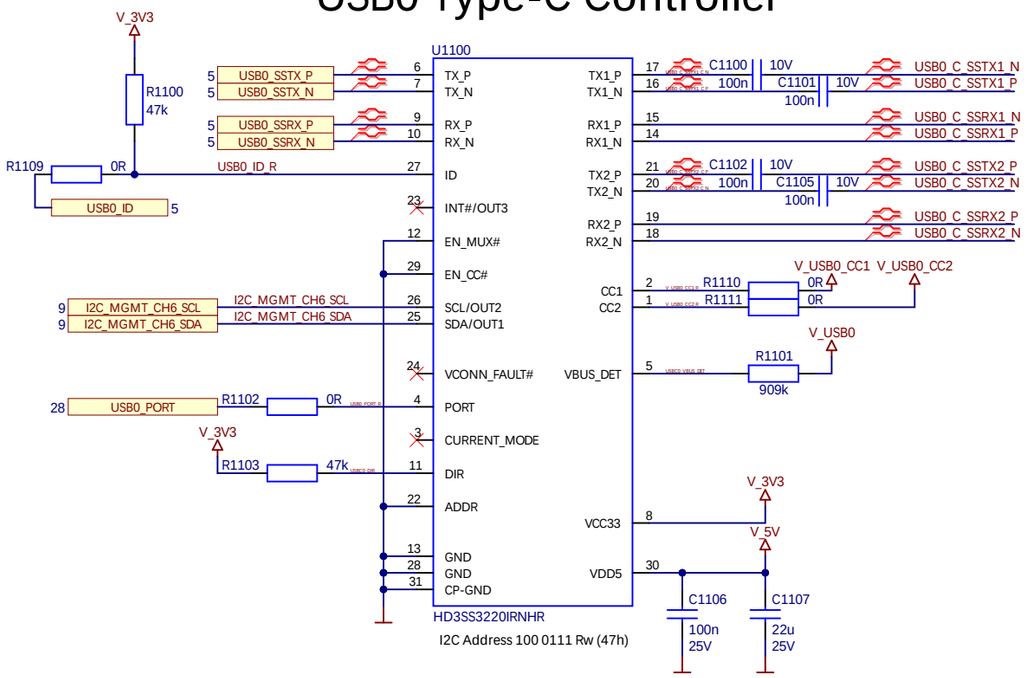


ESD Protection

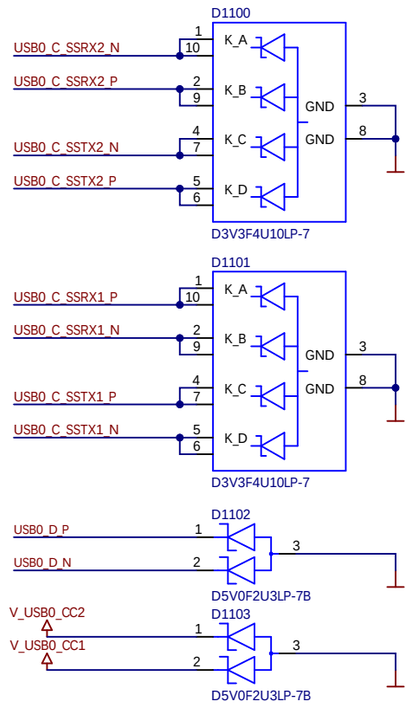


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P10_USB_Config.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 10 / 36

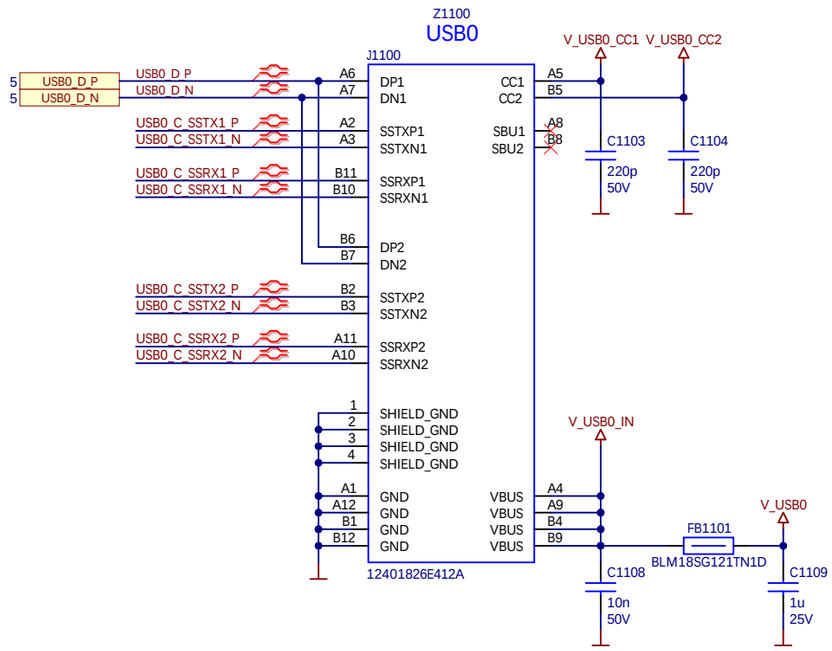
USB0 Type-C Controller



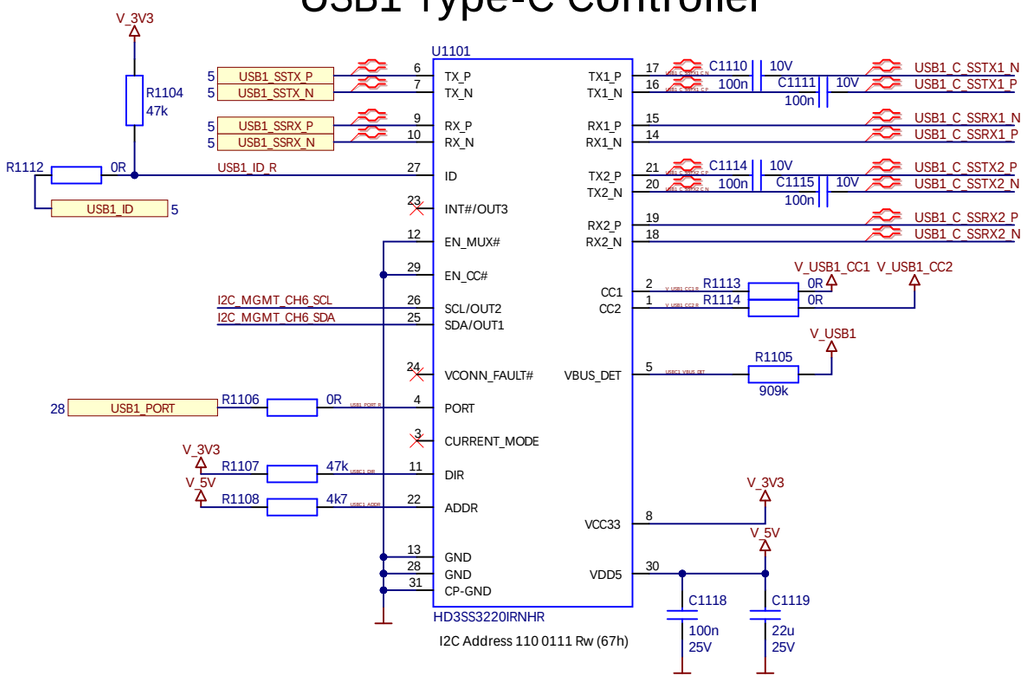
ESD Protection



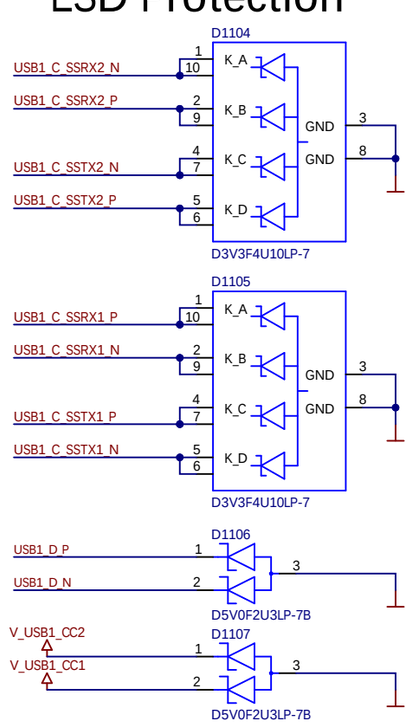
USB0 Type-C Connector



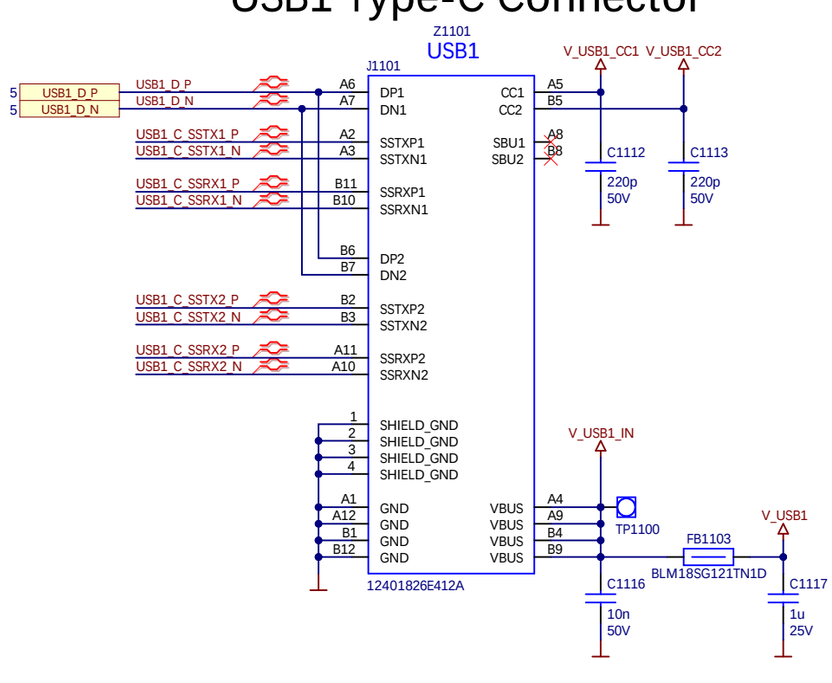
USB1 Type-C Controller



ESD Protection

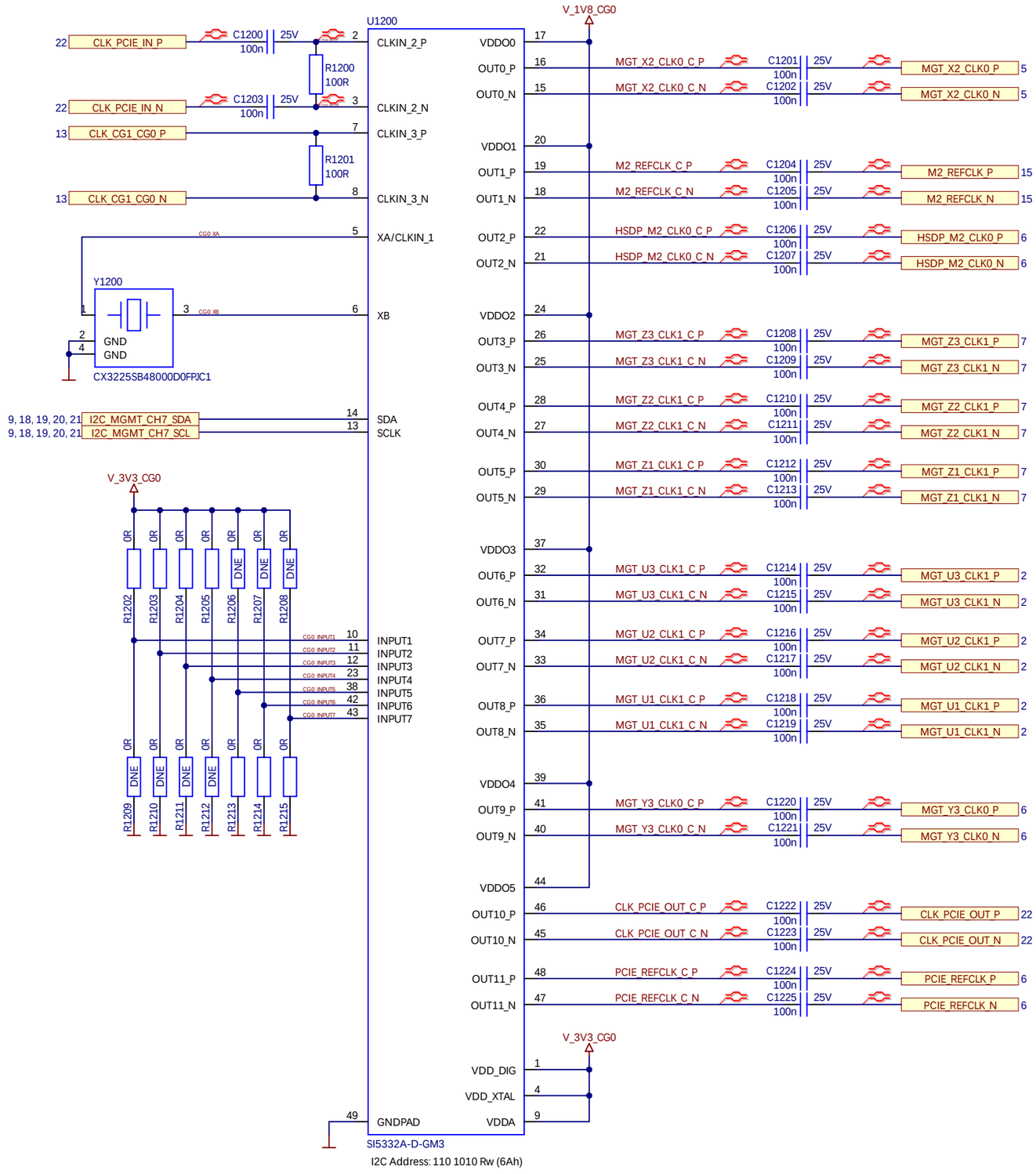


USB1 Type-C Connector



ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P11_USB3.0.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 11 / 36

Clock Generator 0



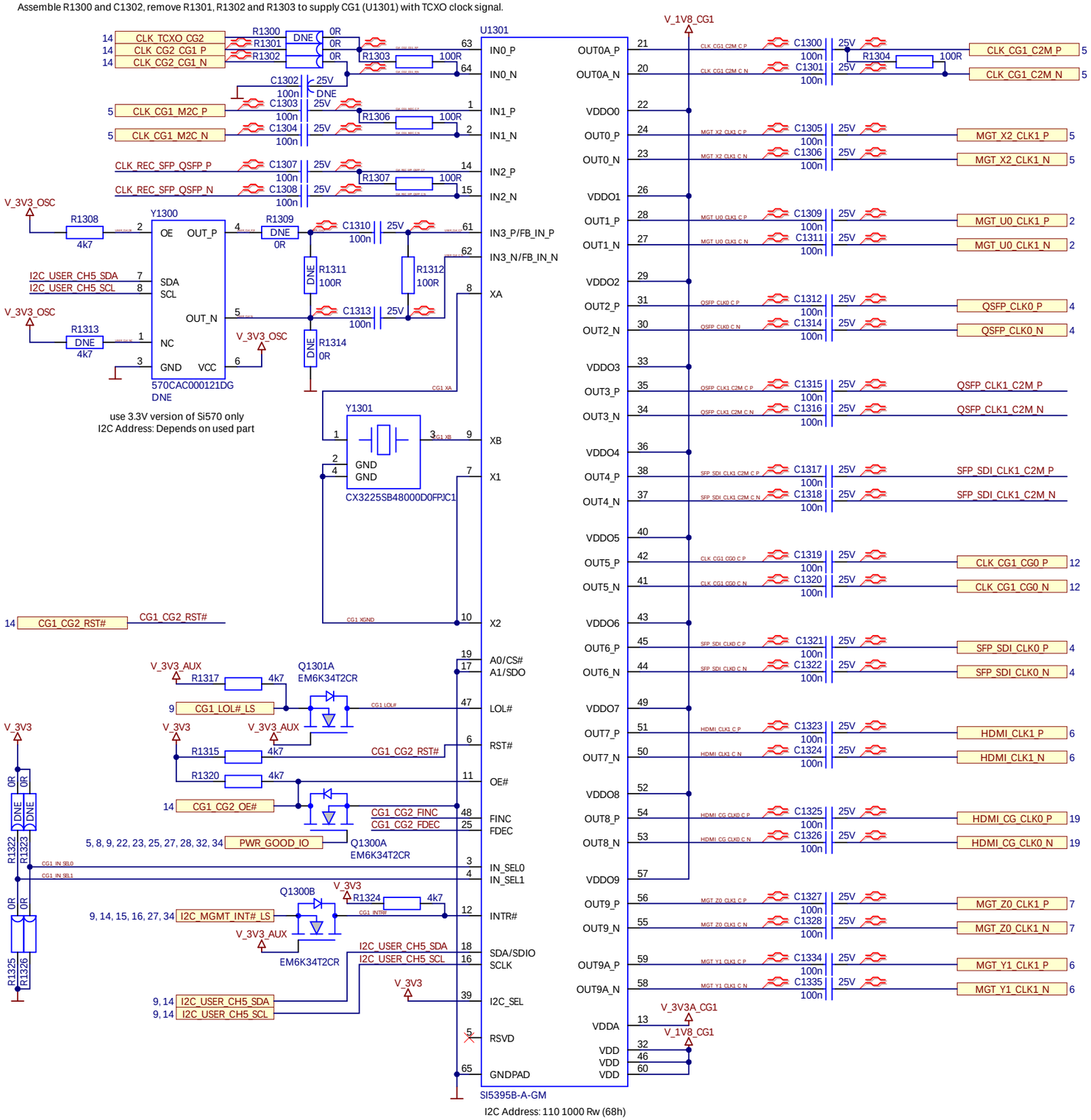
Power Filters

not included in user schematic

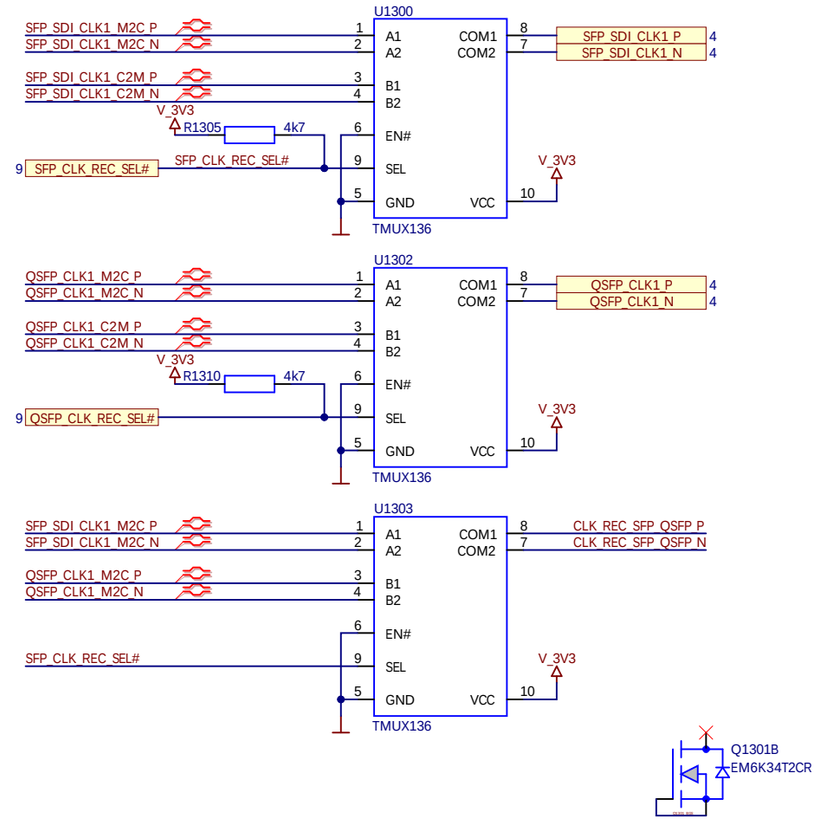
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P12_Oscillator_ClockGenerator.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 12 / 36

Clock Generator 1

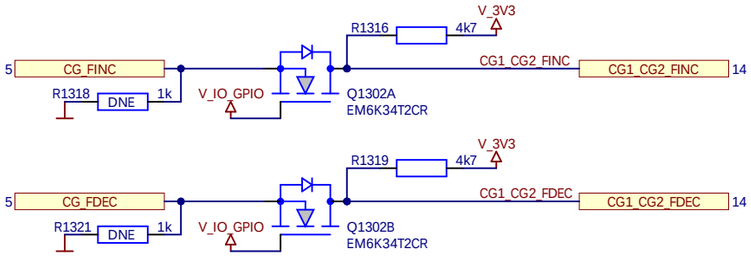
Assemble R1300 and C1302, remove R1301, R1302 and R1303 to supply CG1 (U1301) with TCXO clock signal.



SFP/QSFP Clock Multiplexing



Control Signal Level Shifters

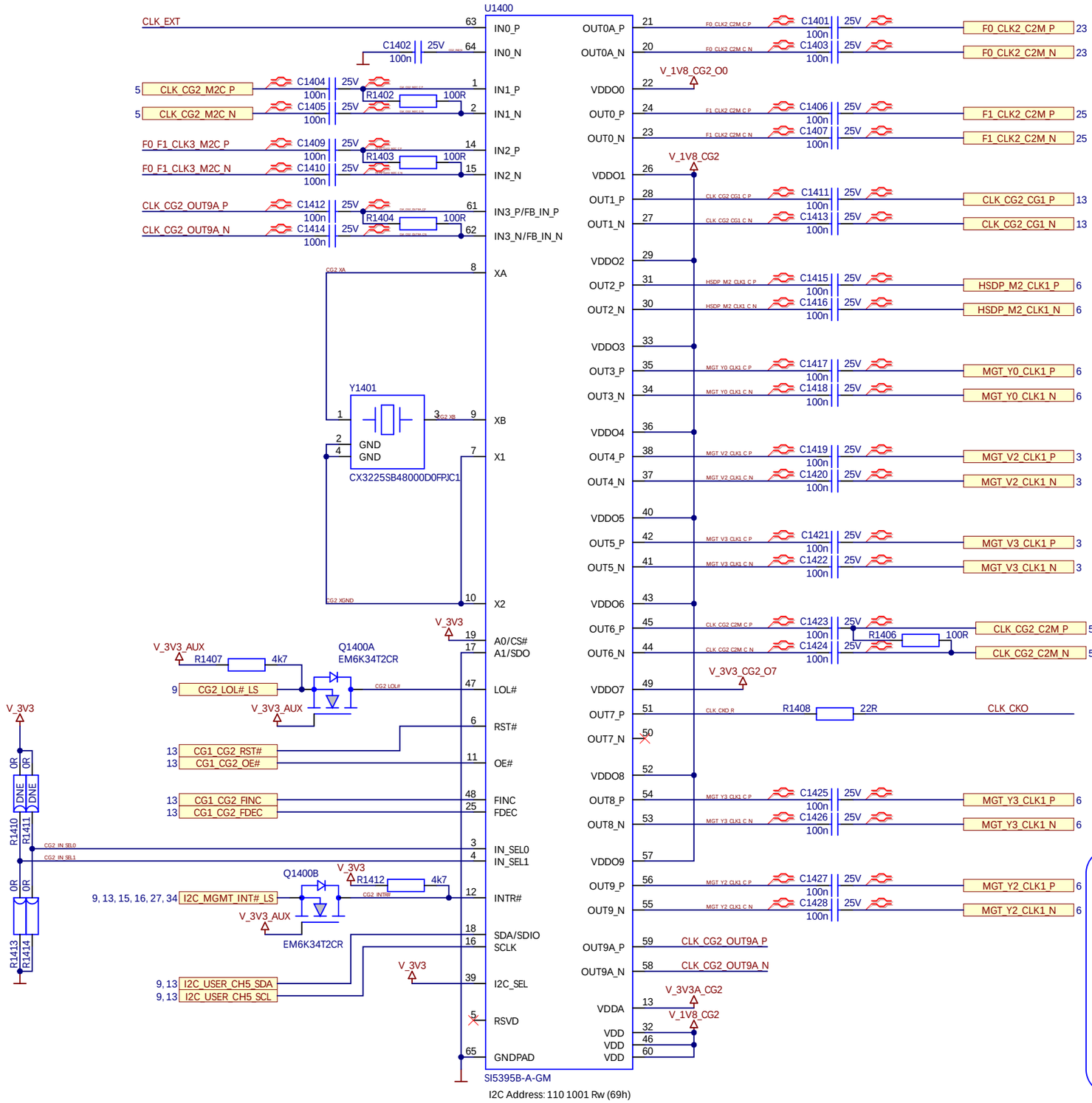


Power Filters

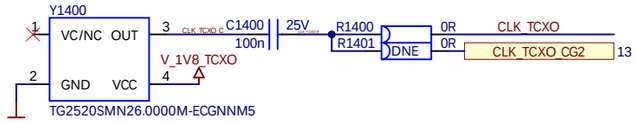
not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P13_Oscillator_ClockGenerator.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 13 / 36

Clock Generator 2

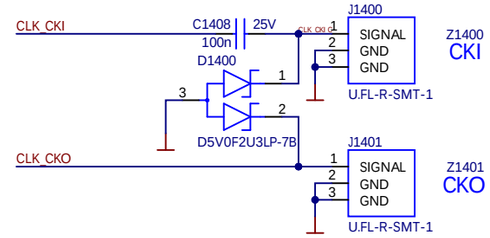


TCXO

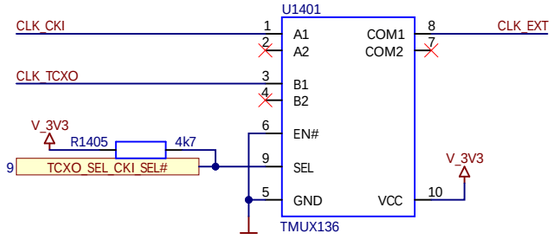


Assemble R1401 and remove R1400 to supply CG1 (U1301) with TCXO clock signal.

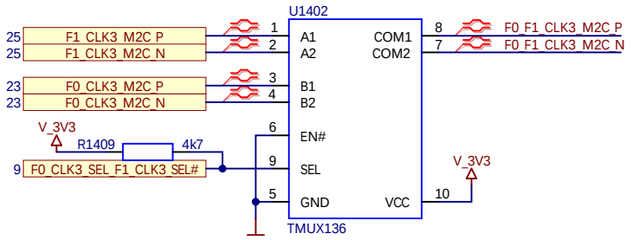
Clock IO Connectors



External Clock Input Multiplexing



FMC CLK3 Clock Multiplexing

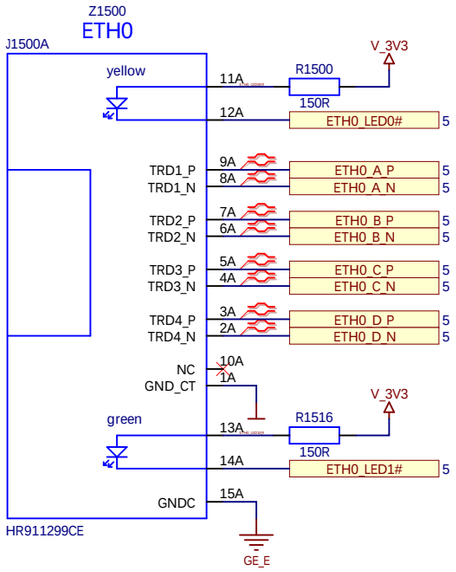


Power Filters

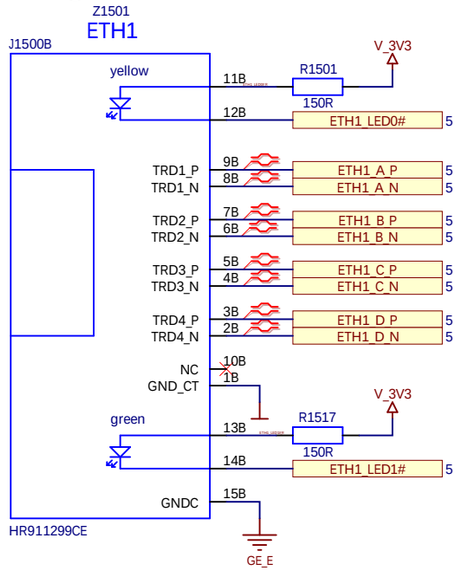
not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P14_Oscillator_ClockGenerator.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 14 / 36

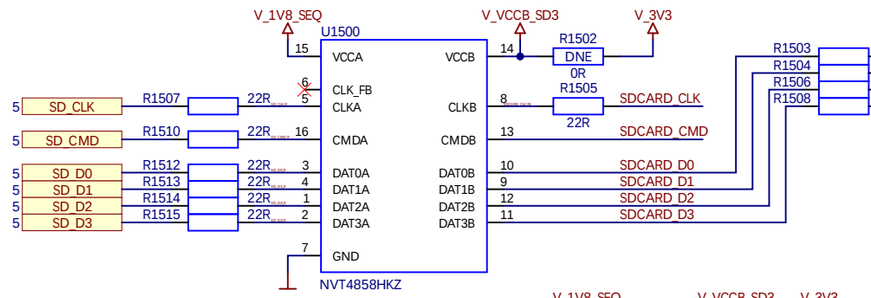
Gigabit Ethernet Port 0



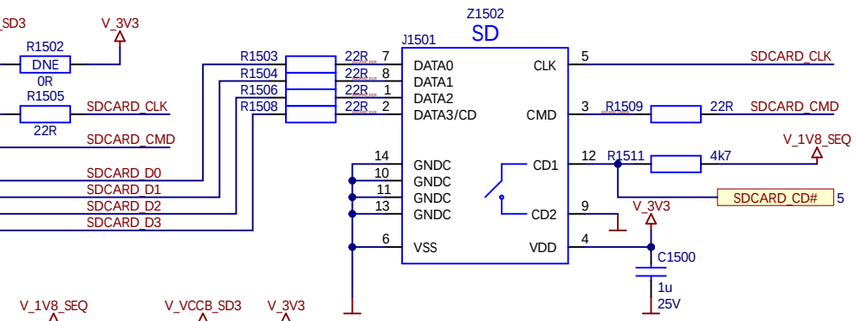
Gigabit Ethernet Port 1

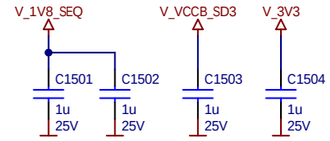


SD3.0 Level Shifter

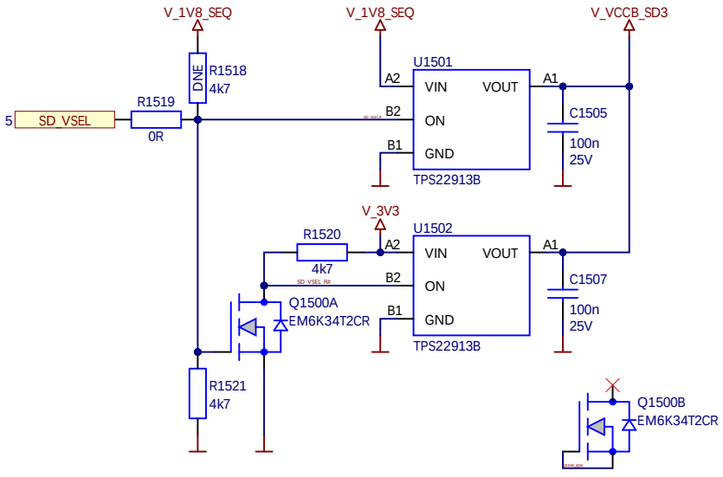


Micro SD-Card Slot

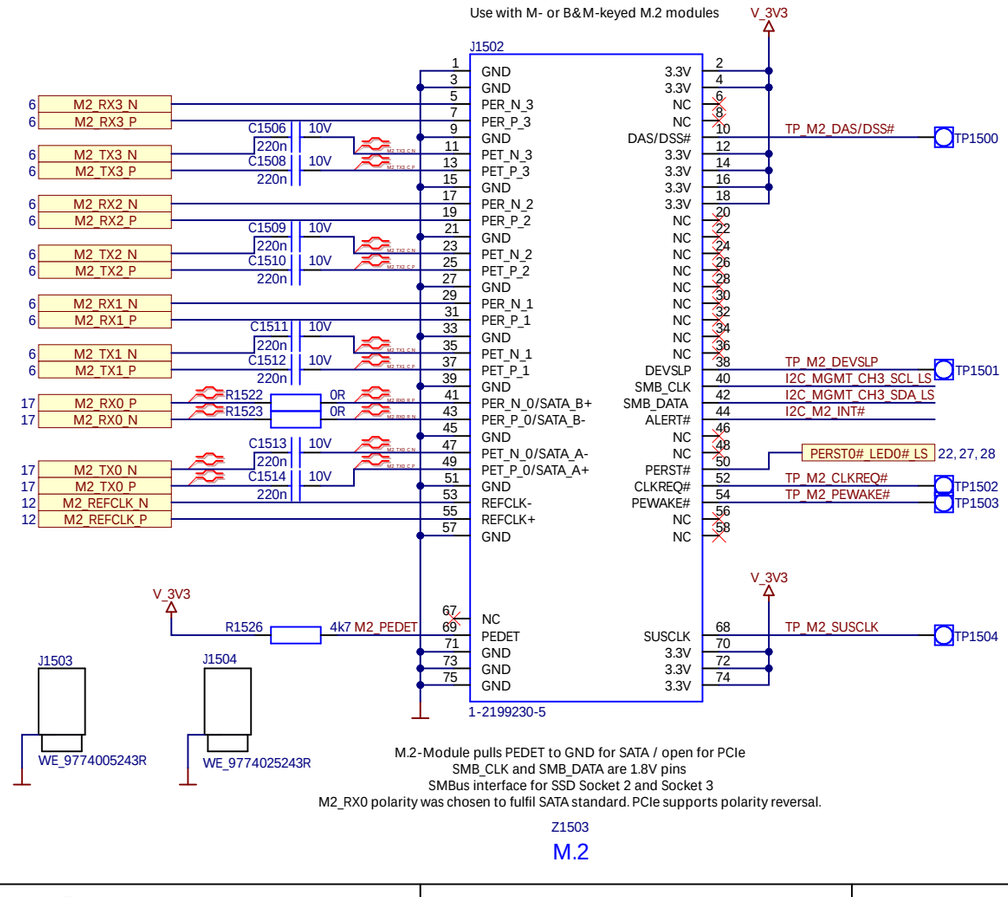




SD3.0 Power Selection



M.2 (NGFF) Socket, Key M



M.2-Module pulls PEDET to GND for SATA / open for PCIe
SMB_CLK and SMB_DATA are 1.8V pins
SMBus interface for SSD Socket 2 and Socket 3
M2_RX0 polarity was chosen to fulfil SATA standard. PCIe supports polarity reversal.

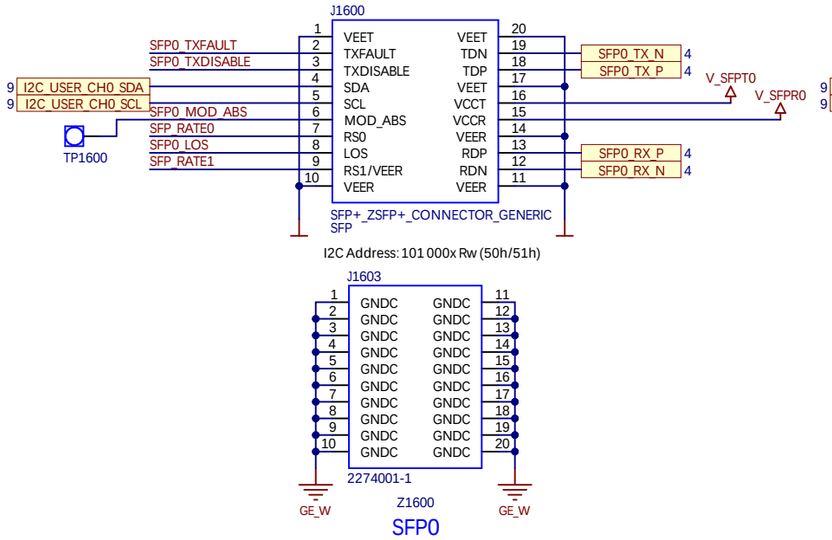
Z1503
M.2

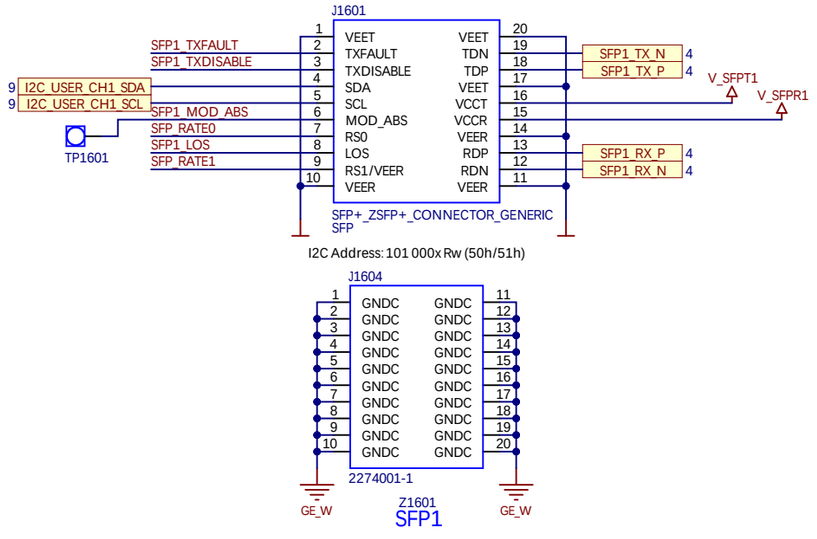
M.2 I2C Level Shifter

M.2 Power Filter

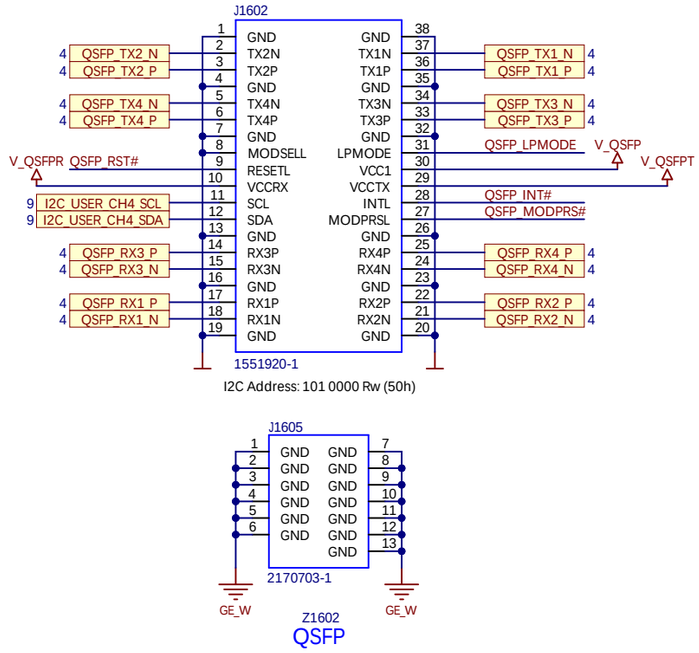
not included in user schematic

SFP28 Connectors and Cages





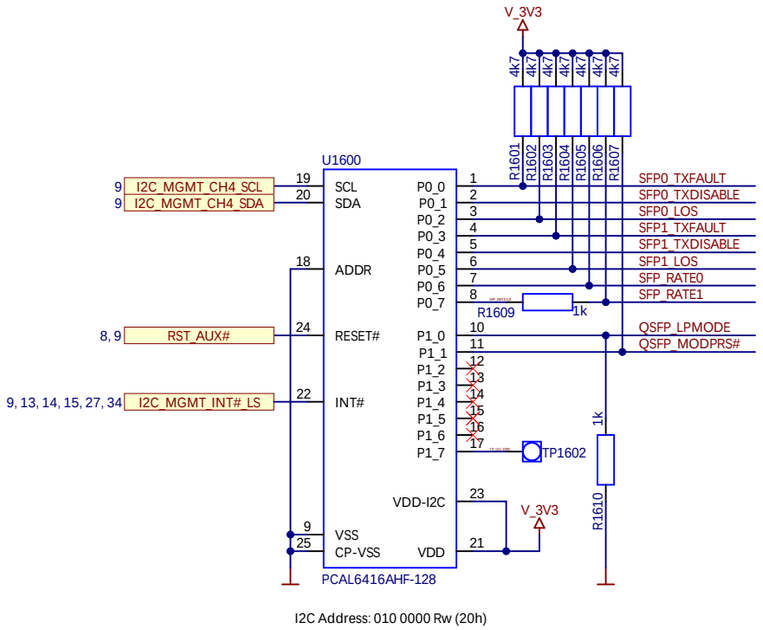
QSFP28 Connector and Cage



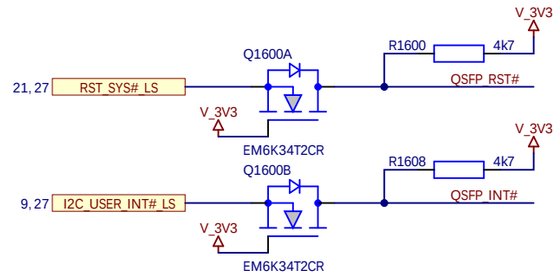
SFP28 Power Filters

not included in user schematic

SFP28/QSFP28 I2C Expander



QSFP28 Level Shifters

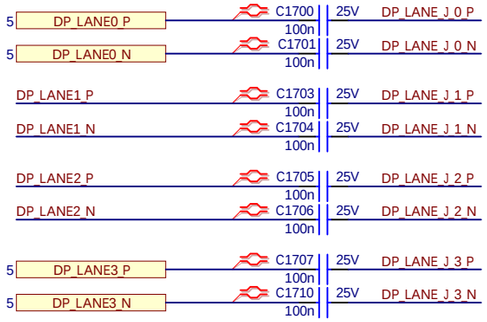


QSFP28 Power Filters

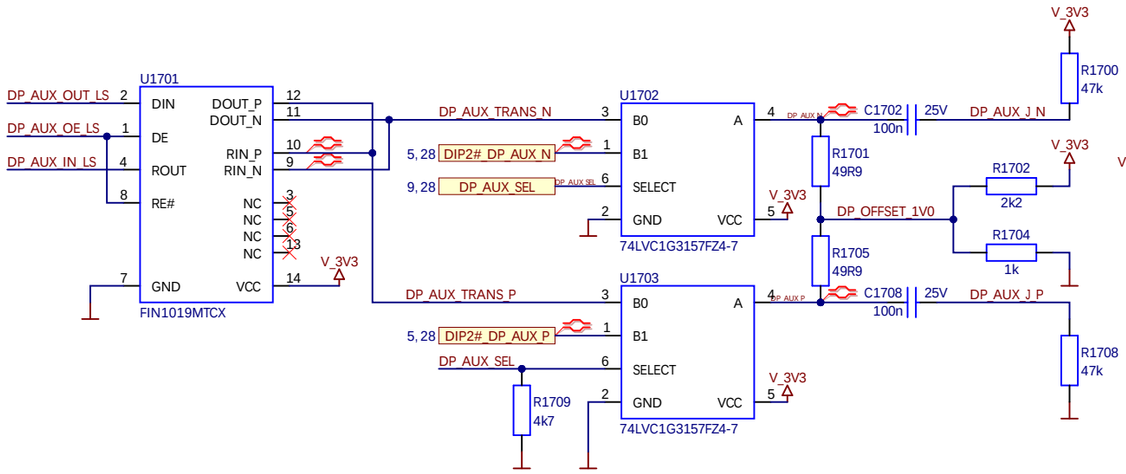
not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P16_SFP_QSFP.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 16 / 36

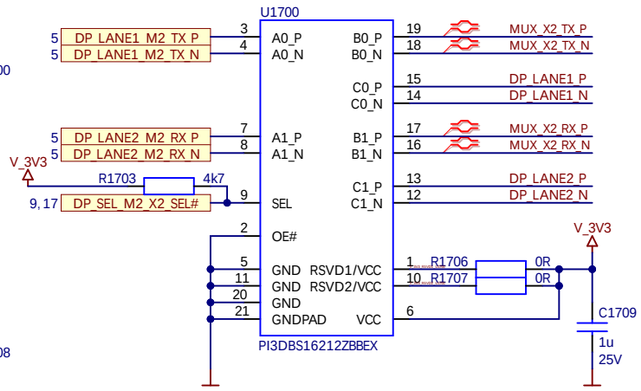
DisplayPort AC Coupling



DisplayPort LVDS Transceiver



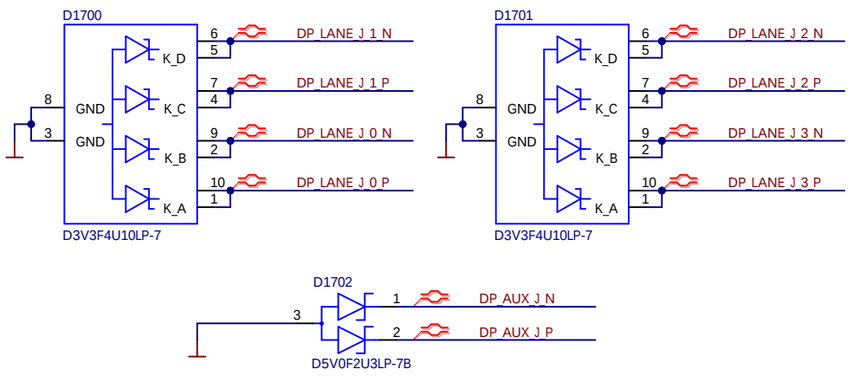
Transceiver Multiplexing



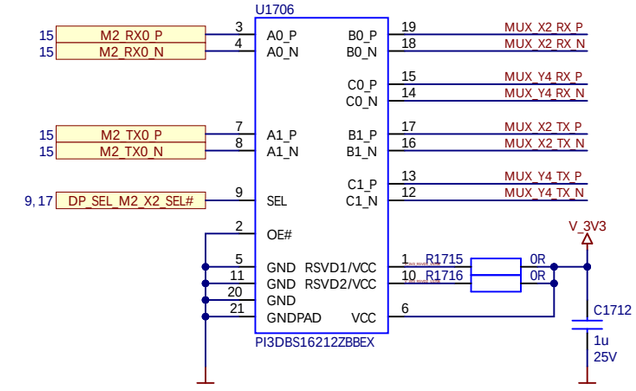
DisplayPort Aux Level Shifter

Mini DisplayPort Connector

DisplayPort ESD Protection

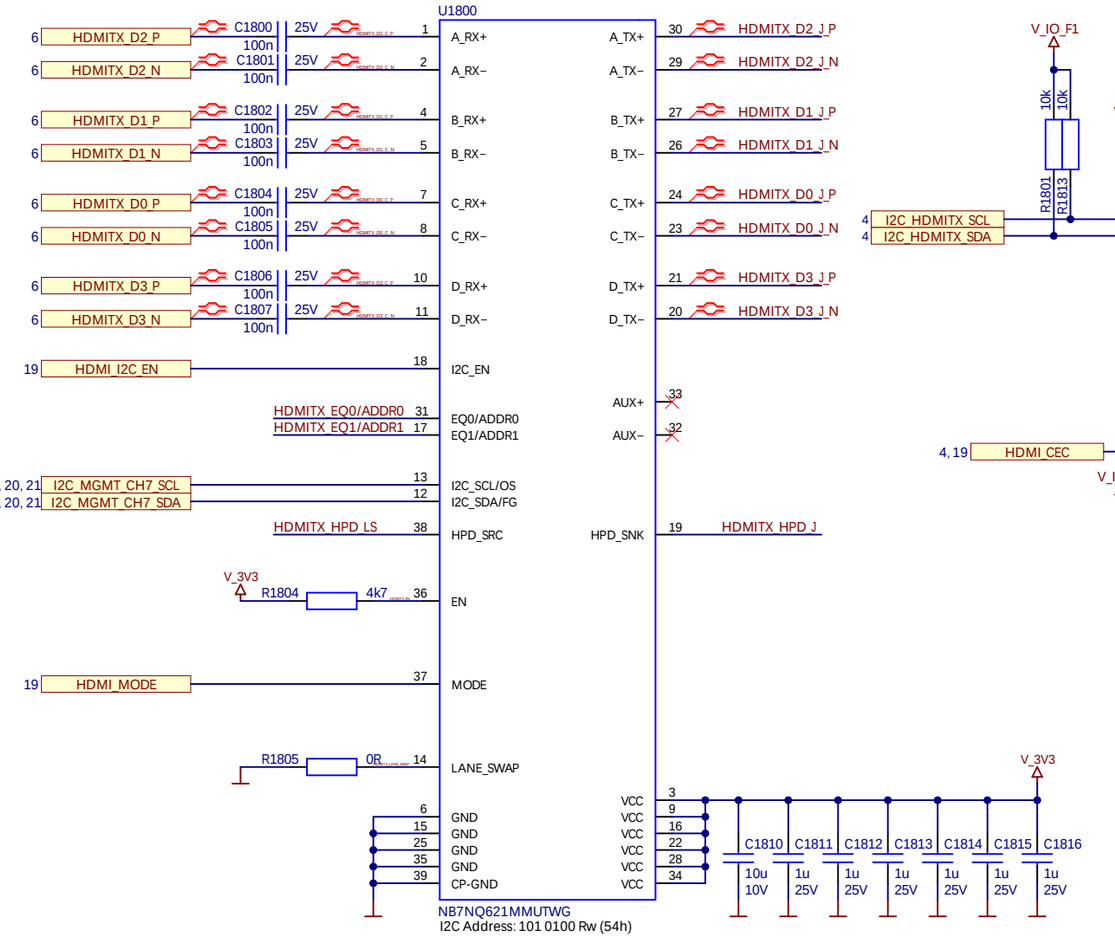


Transceiver Multiplexing

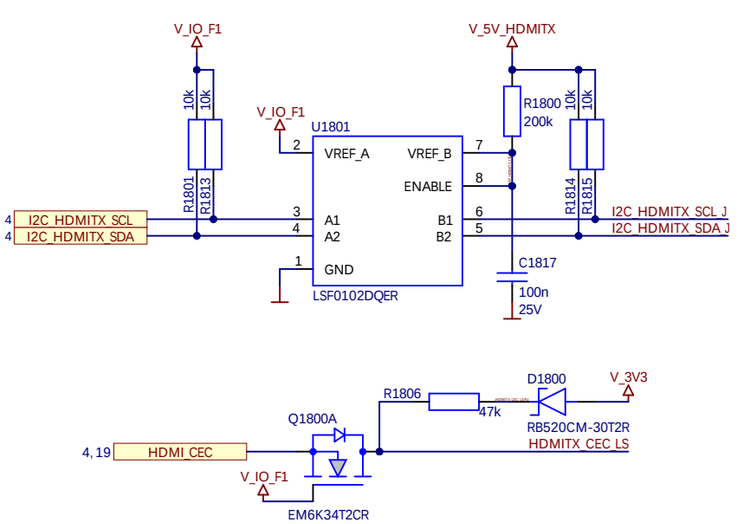


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P17_DisplayPort.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 17 / 36

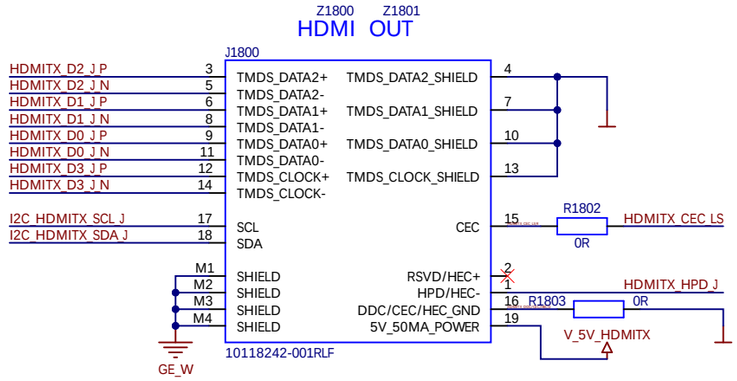
HDMI 2.1 Output Redriver



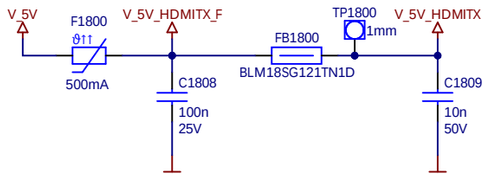
HDMI Output Level Shifter



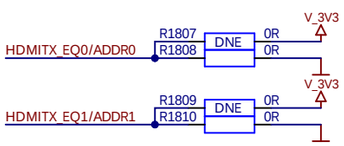
HDMI Output Connector



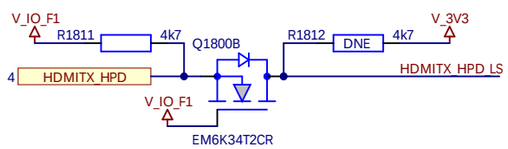
HDMI Output 5V Power



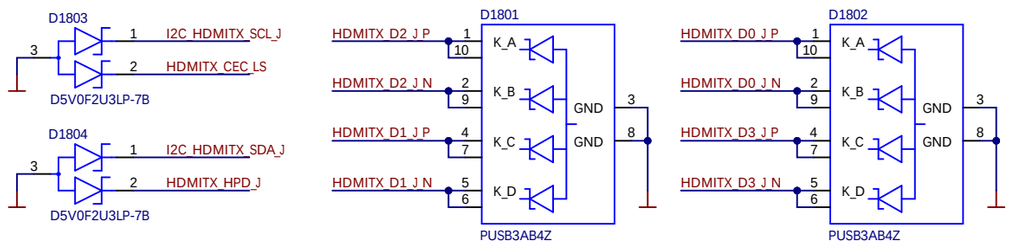
Pin Strapping Options



HDMI Miscellaneous

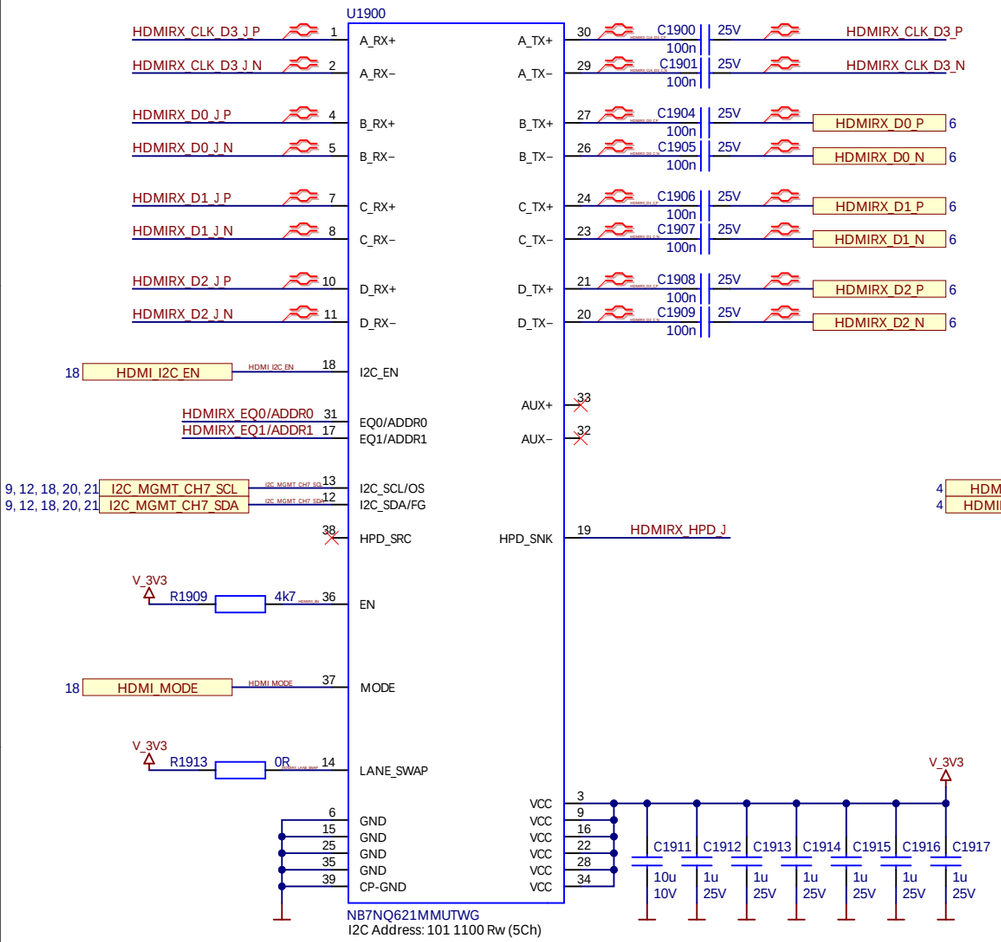


HDMI Output ESD Protection

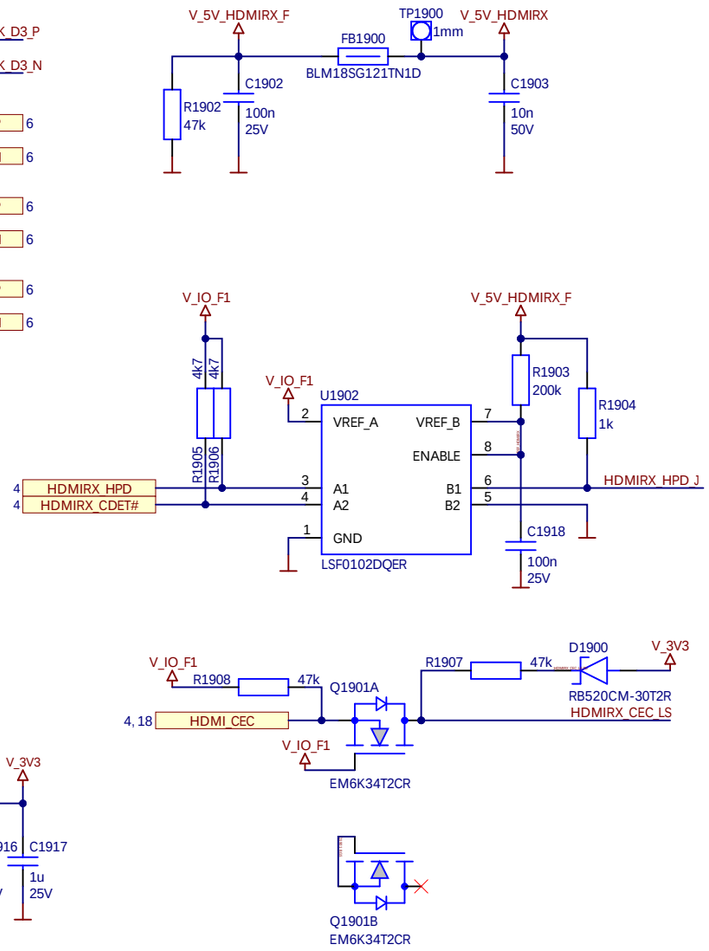


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P18_HDMI_TX.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 18 / 36

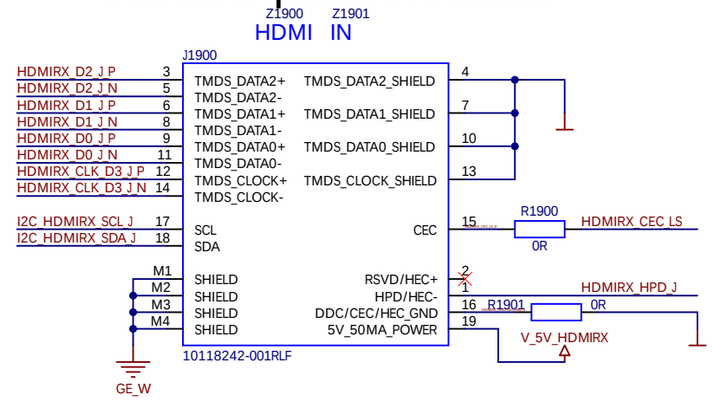
HDMI 2.1 Input Redriver



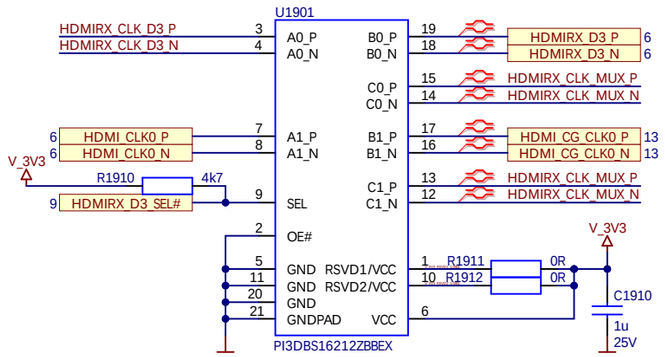
HDMI Input Misc



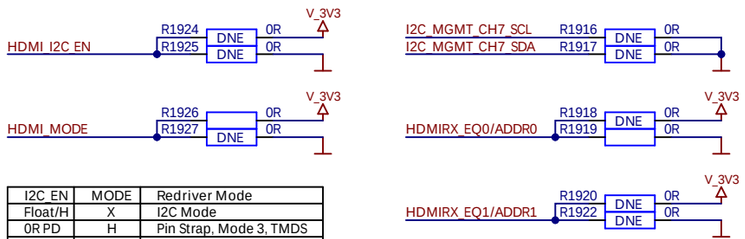
HDMI Input Connector



HDMI Input Clock Mux

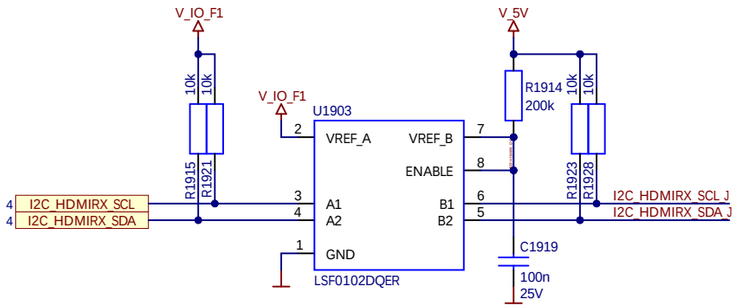


Pin Strapping Options

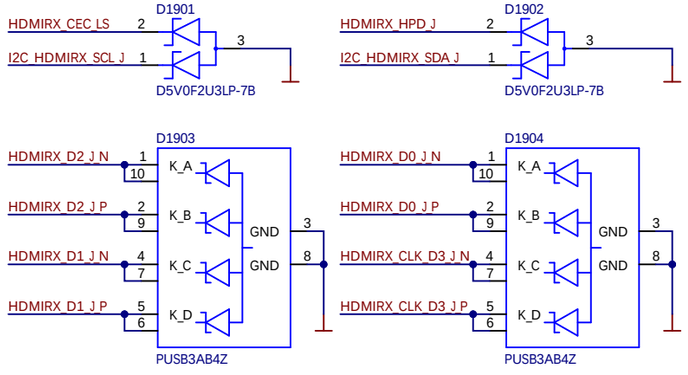


I2C_EN	MODE	Redriver Mode
Float/H	X	I2C Mode
0R PD	H	Pin Strap, Mode 3, TMDS
68k PD	L	Pin Strap, Mode 4, FRL
See NB7NQ621M datasheet for more details		

HDMI Input I2C Level Shifter

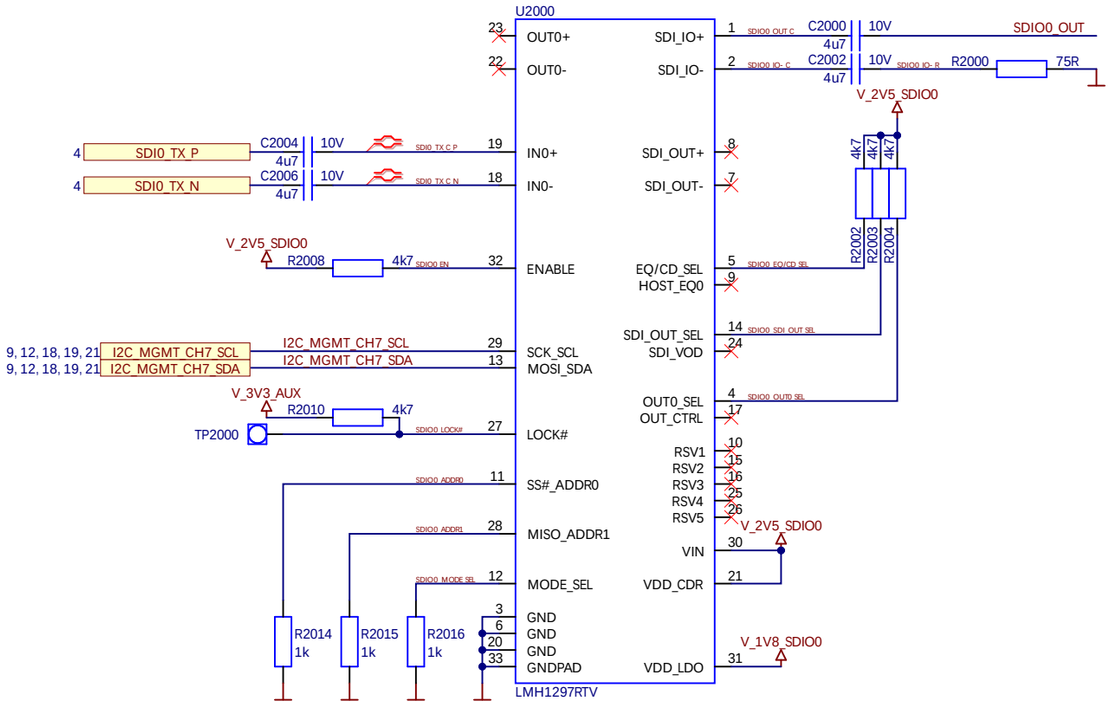


HDMI Input ESD Protection



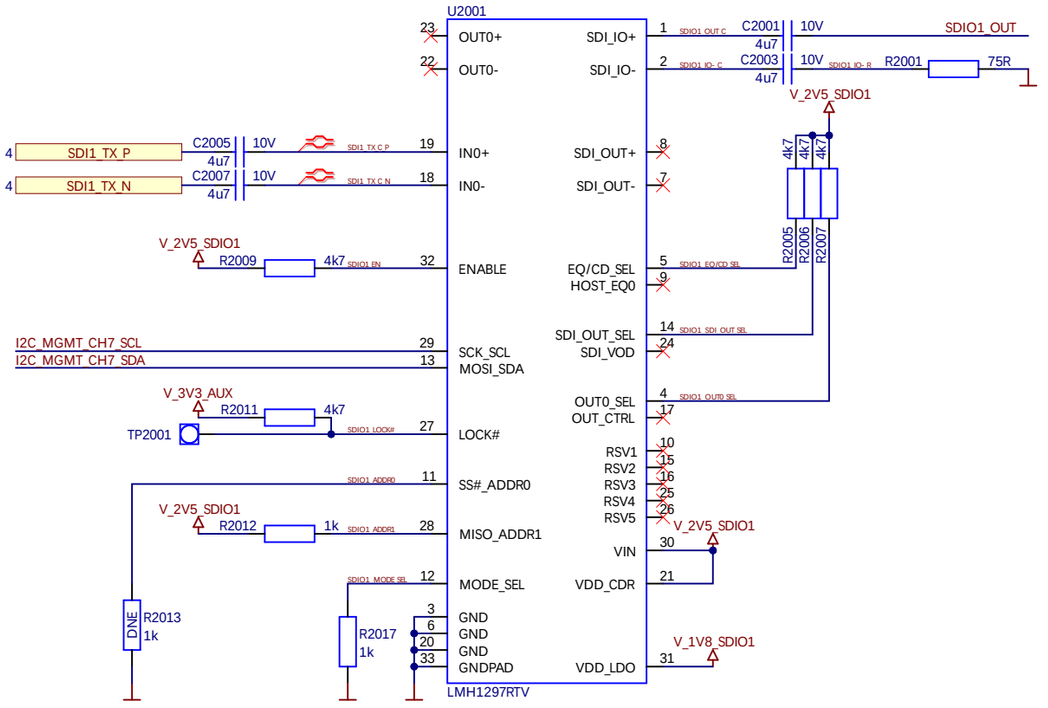
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P19_HDMI_RX.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 19 / 36

SDI0 Output



I2C Address: 010 1101 Rw (2Dh)

SDI1 Output

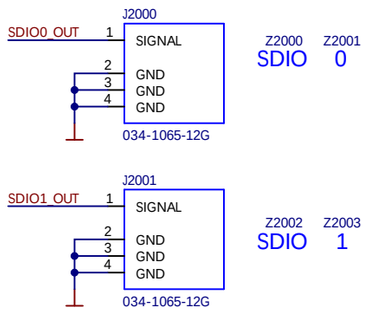


I2C Address: 011 1000 Rw (38h)

SDI Output Power Filters

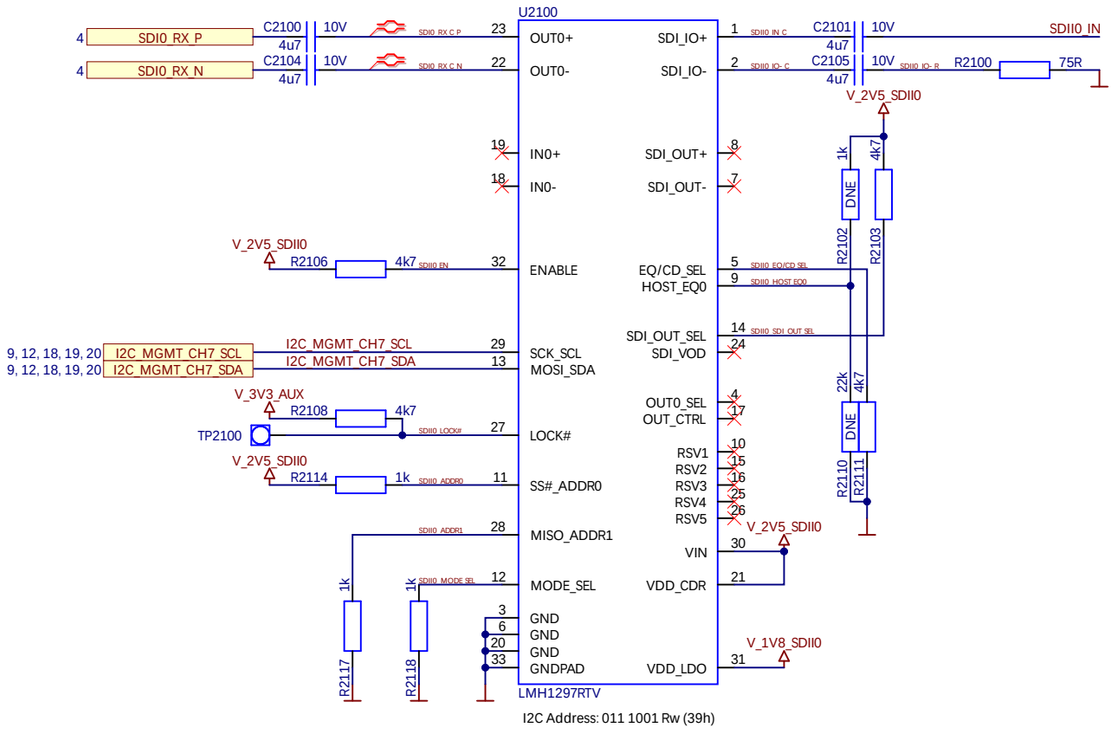
not included in user schematic

SDI Output Connectors



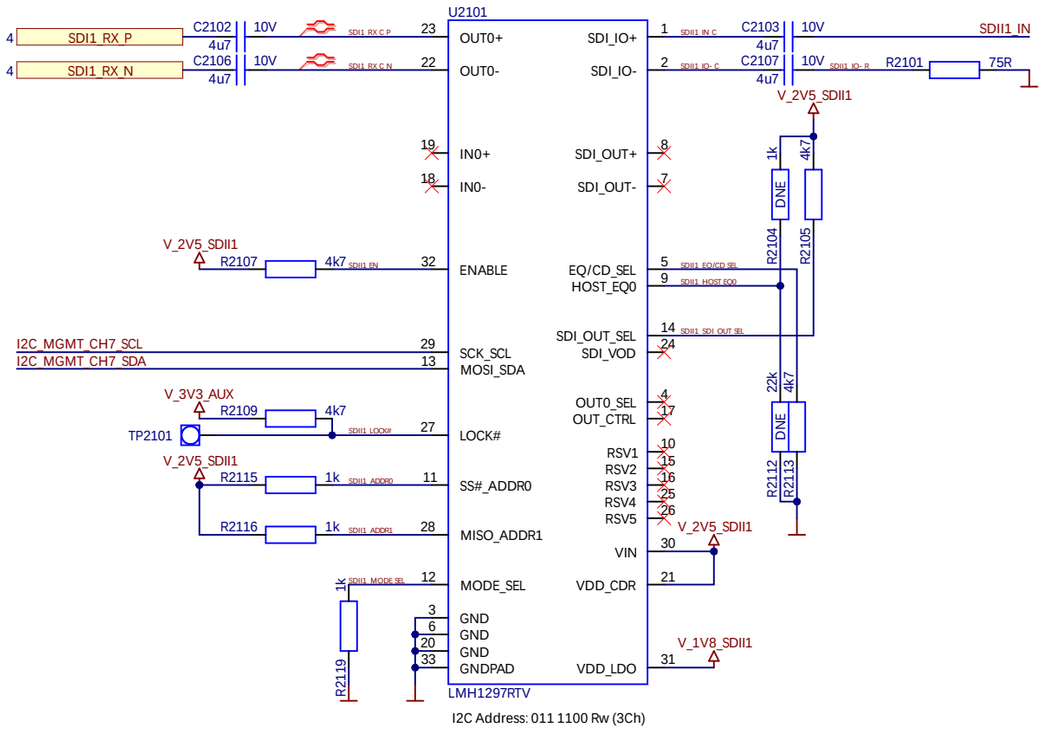
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P20_SDI_TX.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 20 / 36

SDI0 Input



I2C Address: 011 1001 Rw (39h)

SDI1 Input

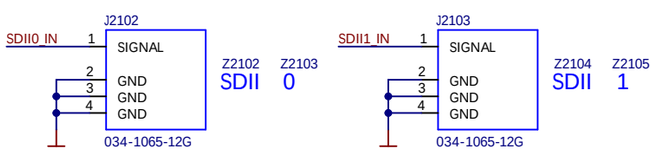


I2C Address: 011 1100 Rw (3Ch)

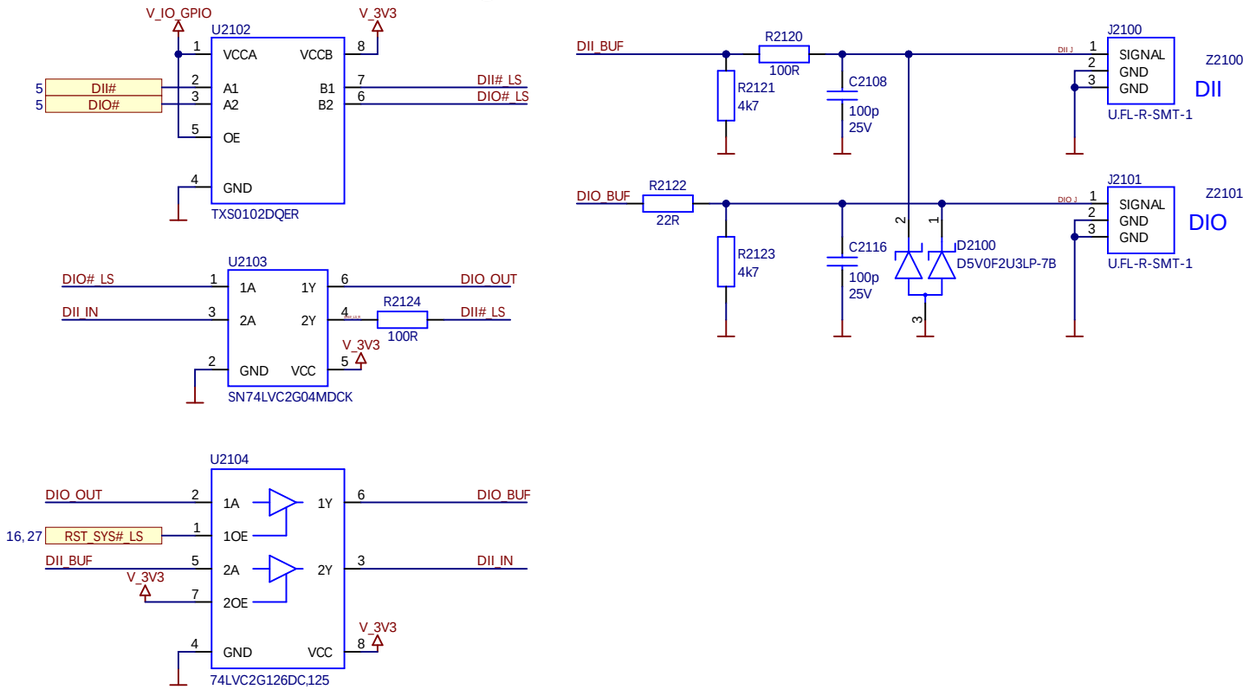
SDI Input Power Filter

not included in user schematic

SDI Input Connectors

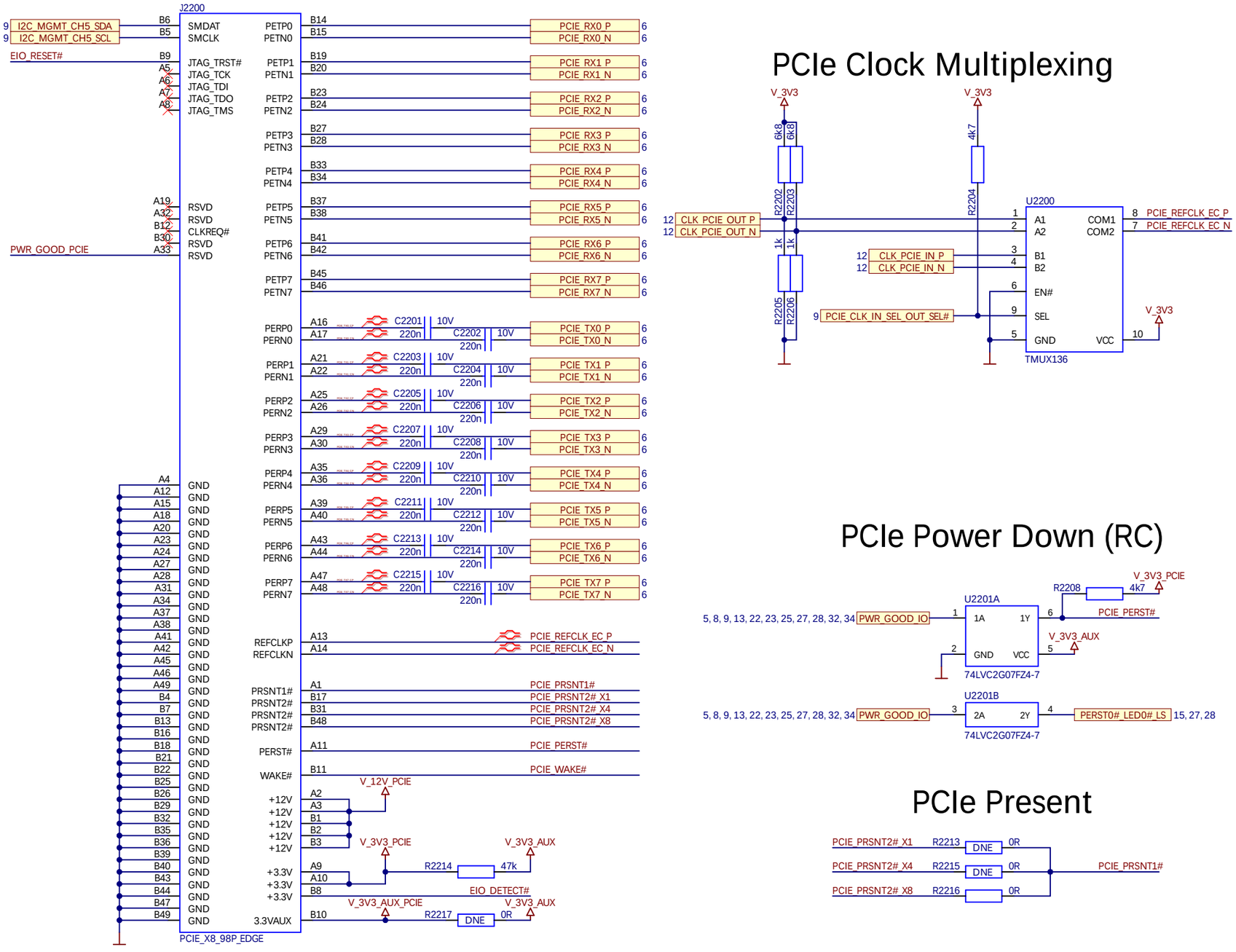


Digital IO Connectors



ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P21_SDI_RX_UFL.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 21 / 36

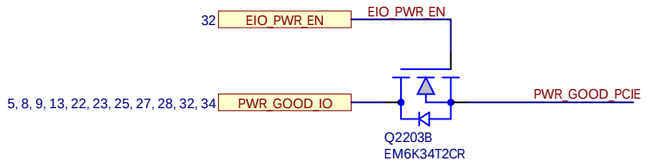
PCIe x8 Edge Connector



PCIe Clock Multiplexing

PCIe Power Down (RC)

PCIe Present

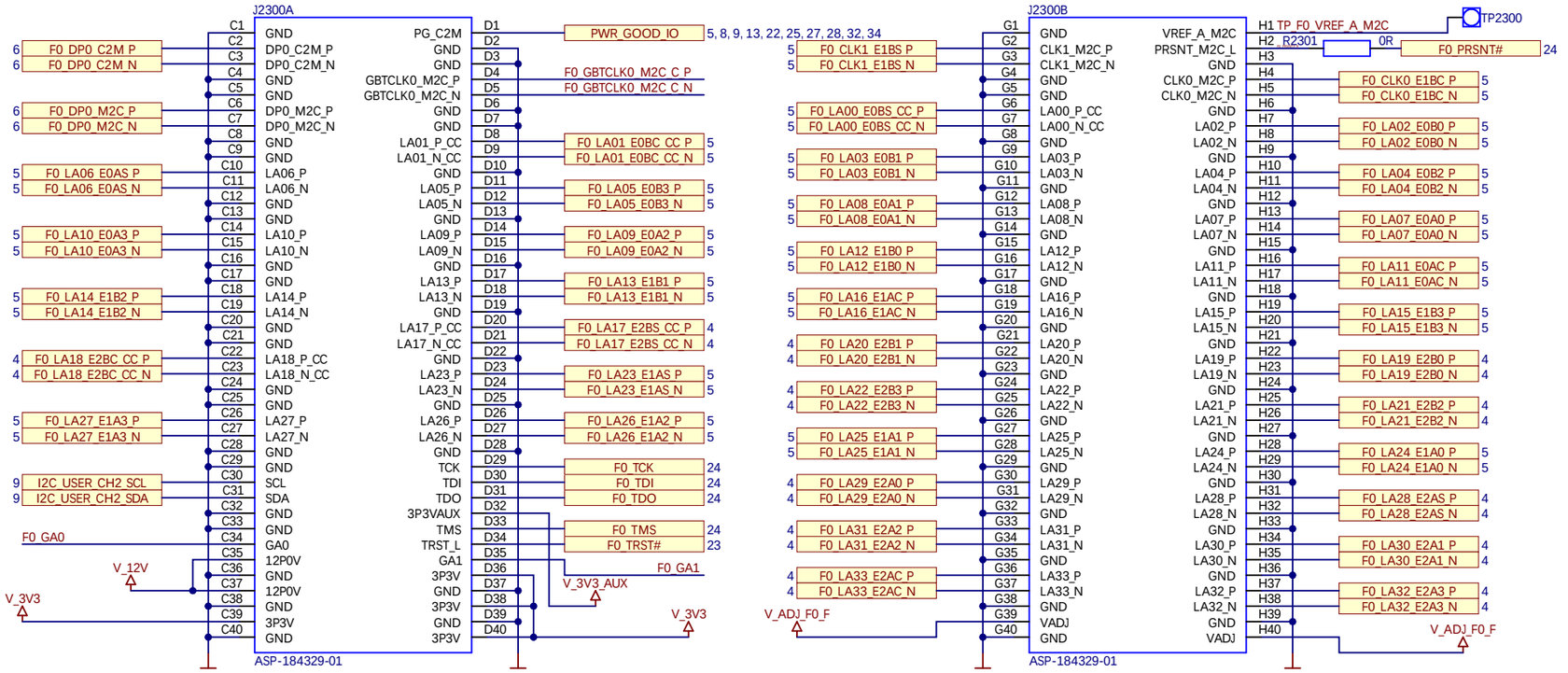


PCIe Power Input & Edge IO Power

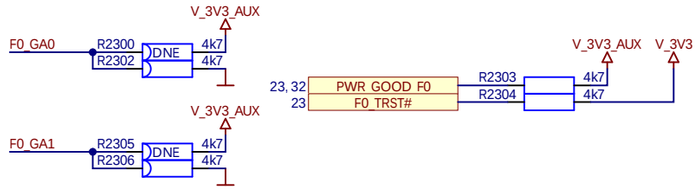
not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P22_PCIe_Edge.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 22 / 36

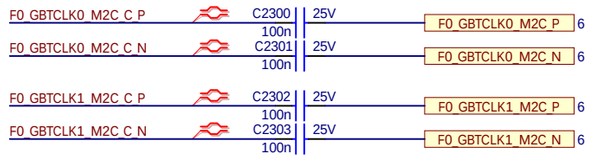
FMC HSPC Connector 0 A-K



FMC 0 Micellaneous



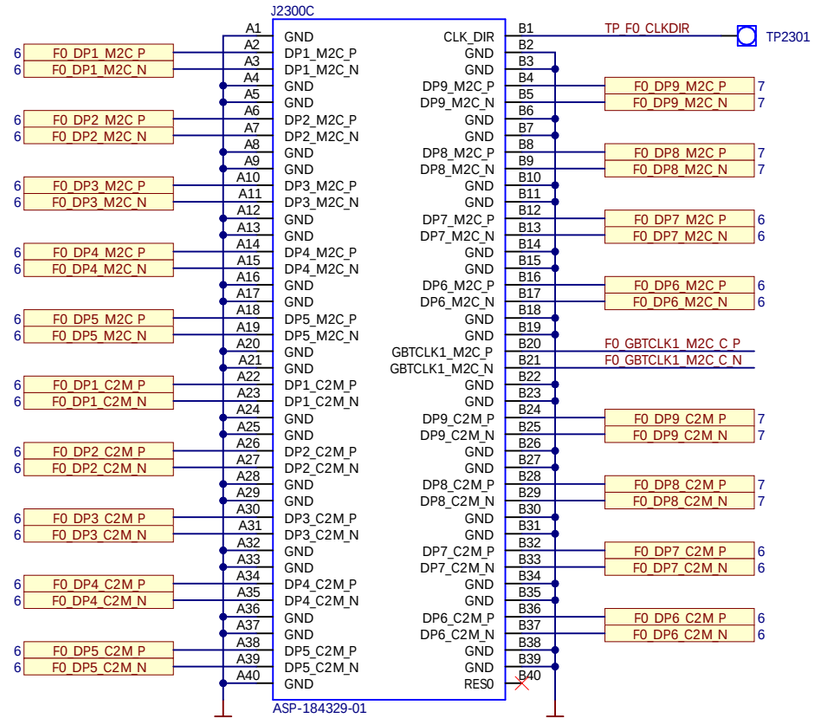
FMC 0 GBTCLK AC Coupling

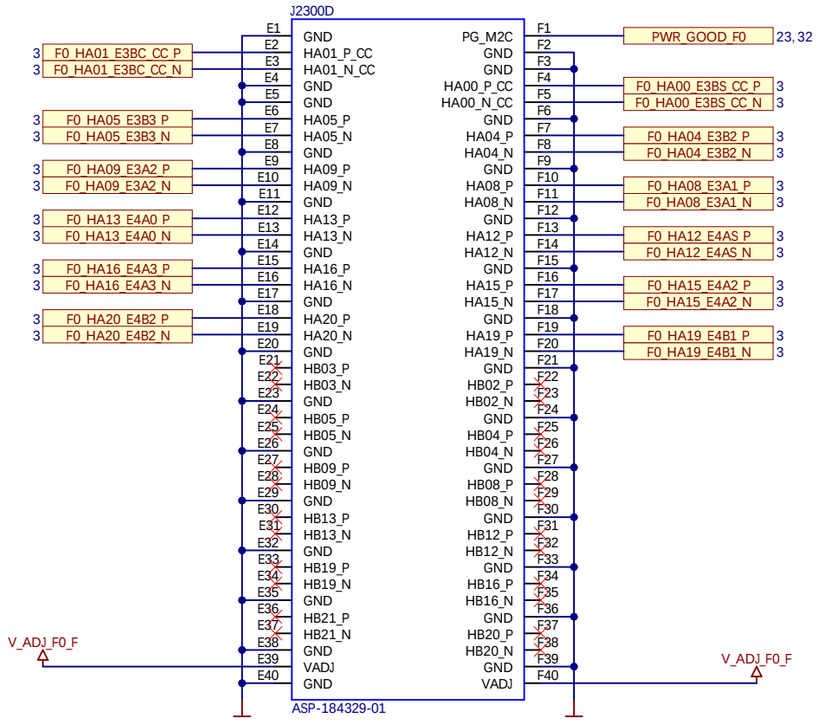


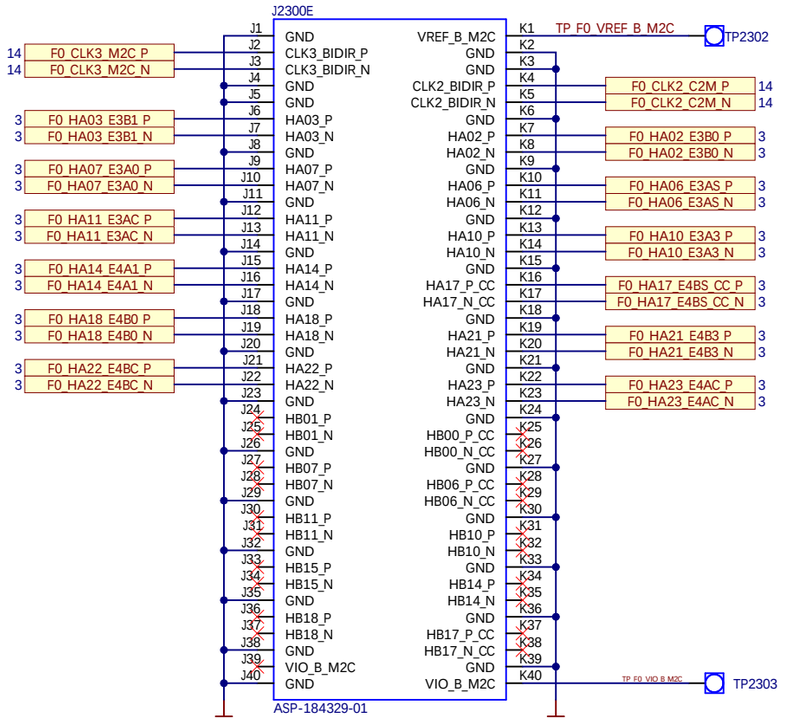
FMC 0 Power Filter

not included in user schematic



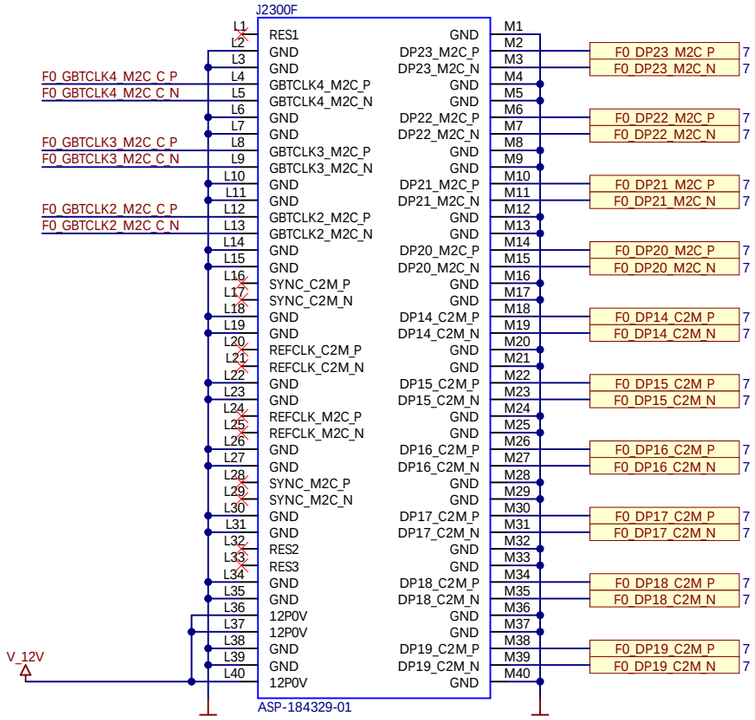


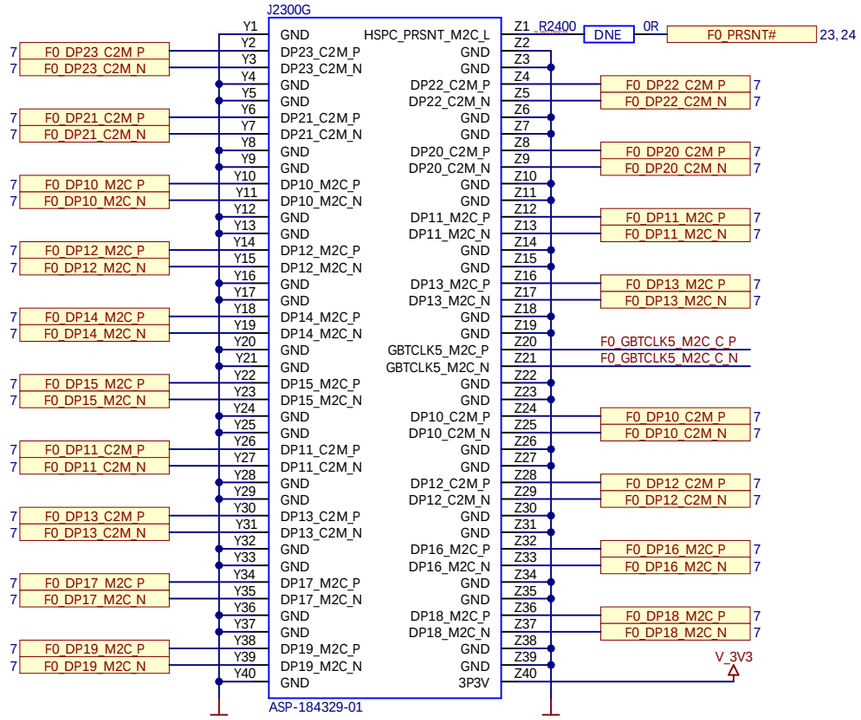




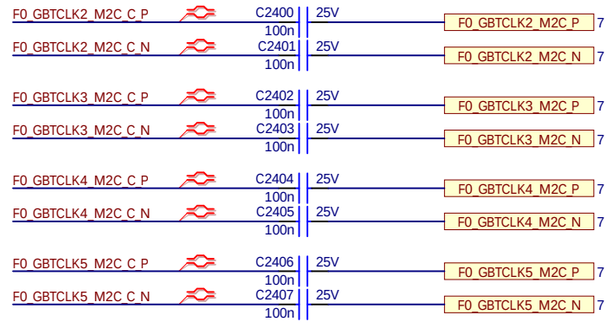
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P23_F0_1.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 23 / 36

FMC HSPC Connector 0 M, L, Y, Z

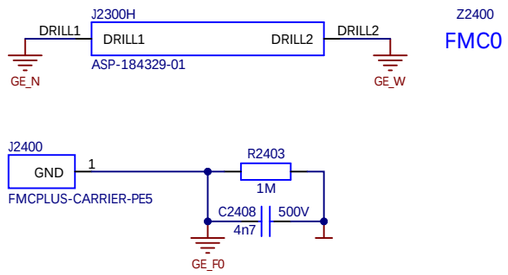




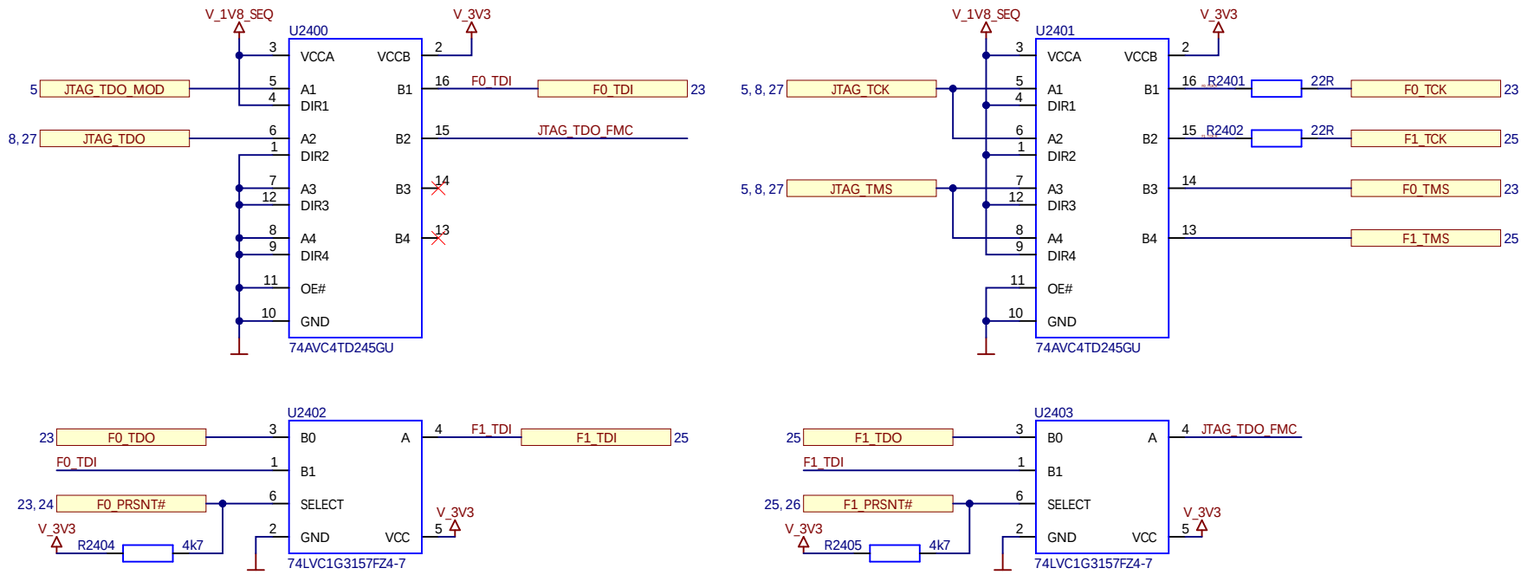
FMC 0 GBTCLK AC Coupling



FMC HSPC Connector 0 Mechanics

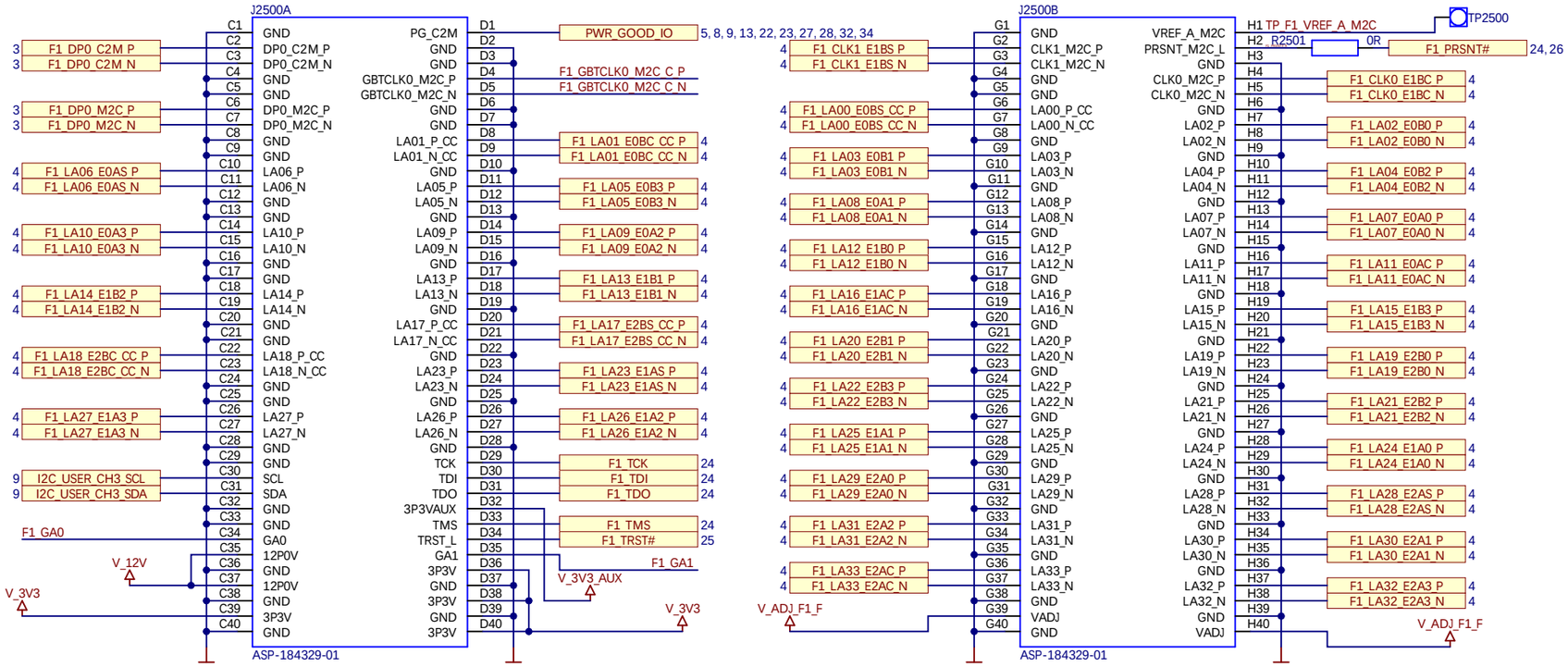


FMC JTAG Chain

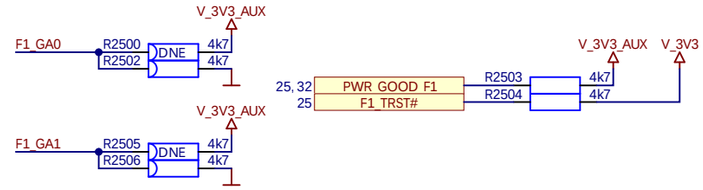


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P24_F0_2.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 24 / 36

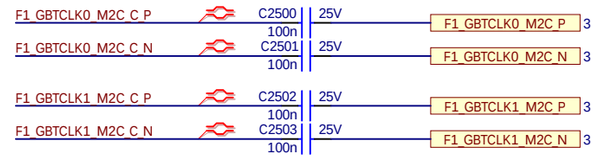
FMC HSPC Connector 1 A-K



FMC 1 Micellaneous



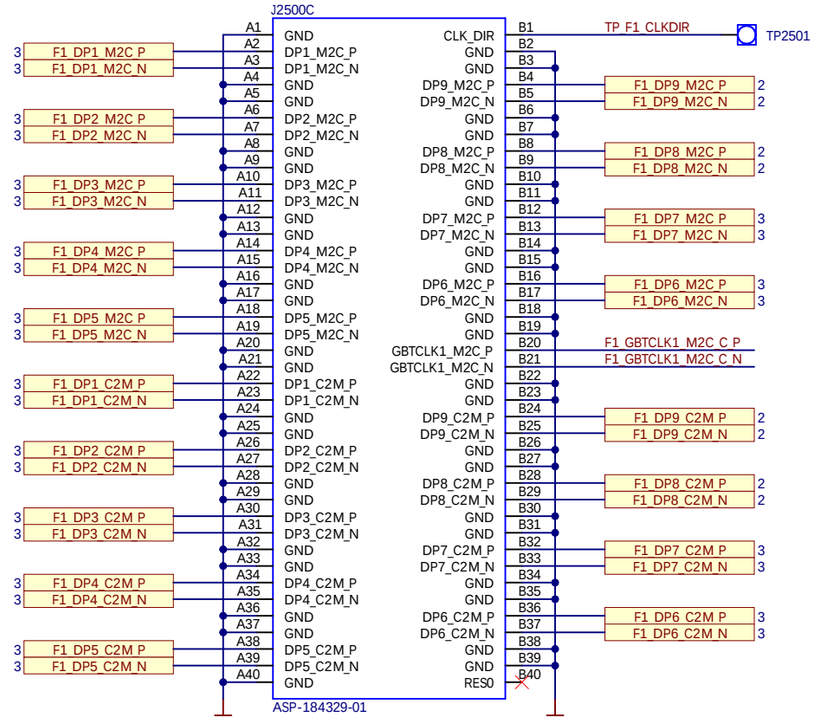
FMC 1 GBTCLK AC Coupling

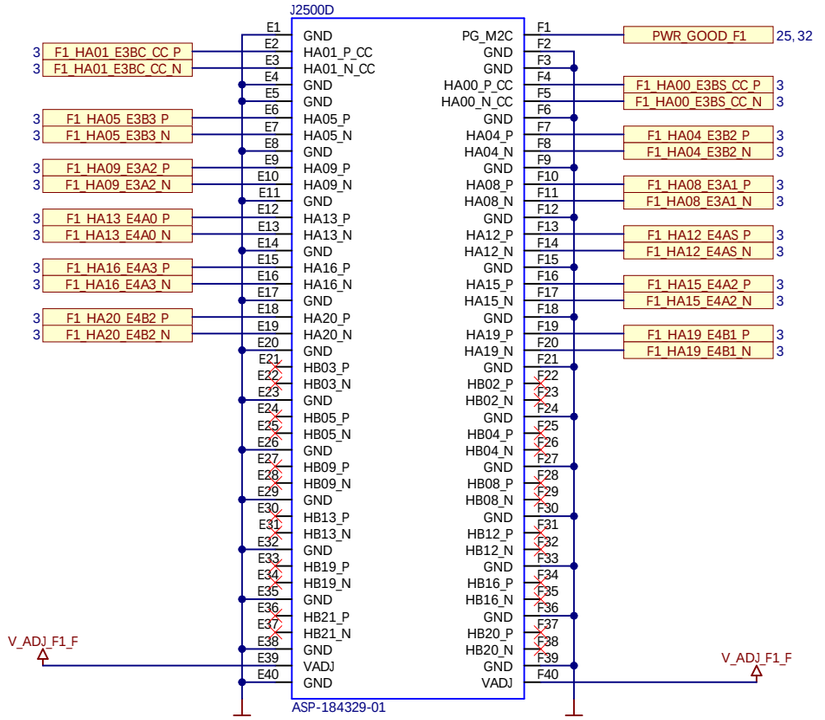


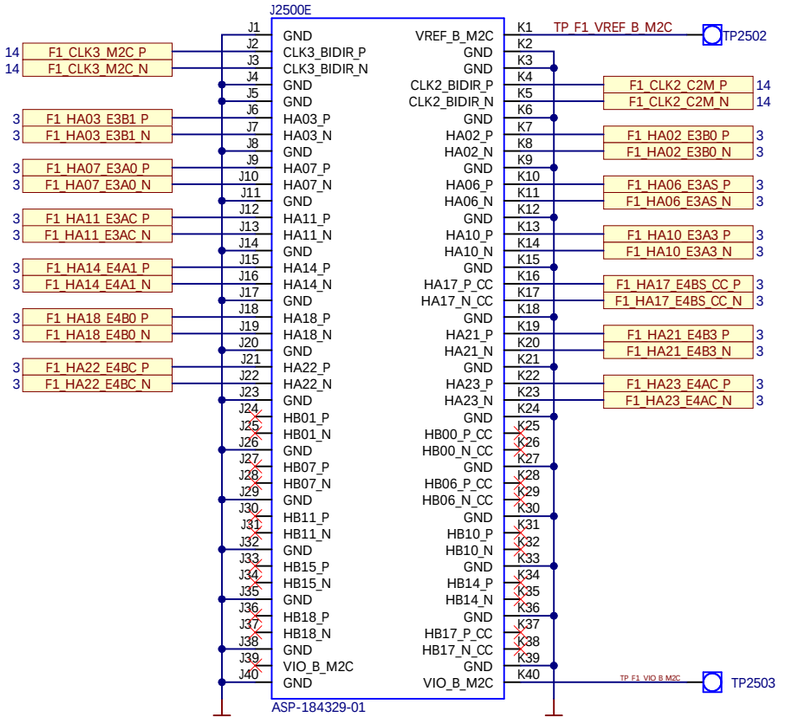
FMC 1 Power Filter

not included in user schematic



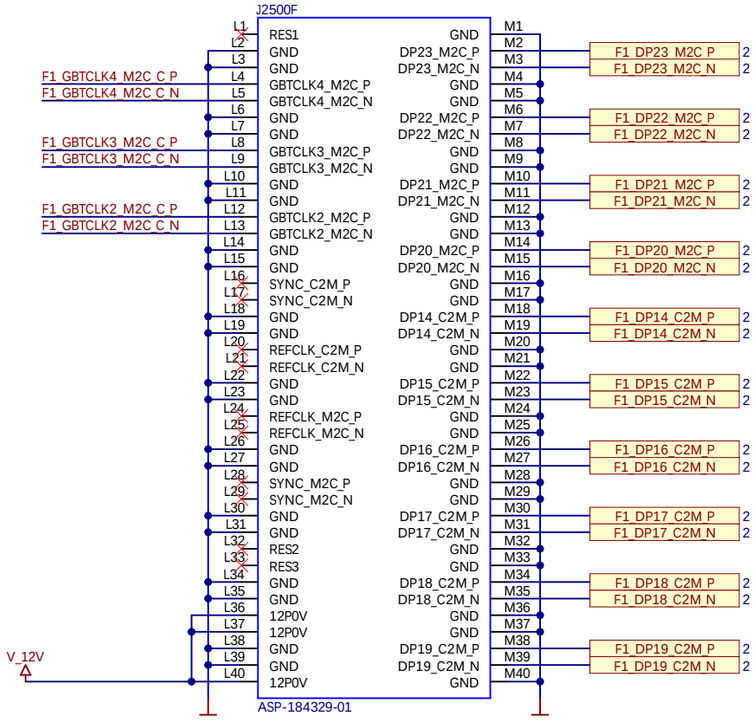


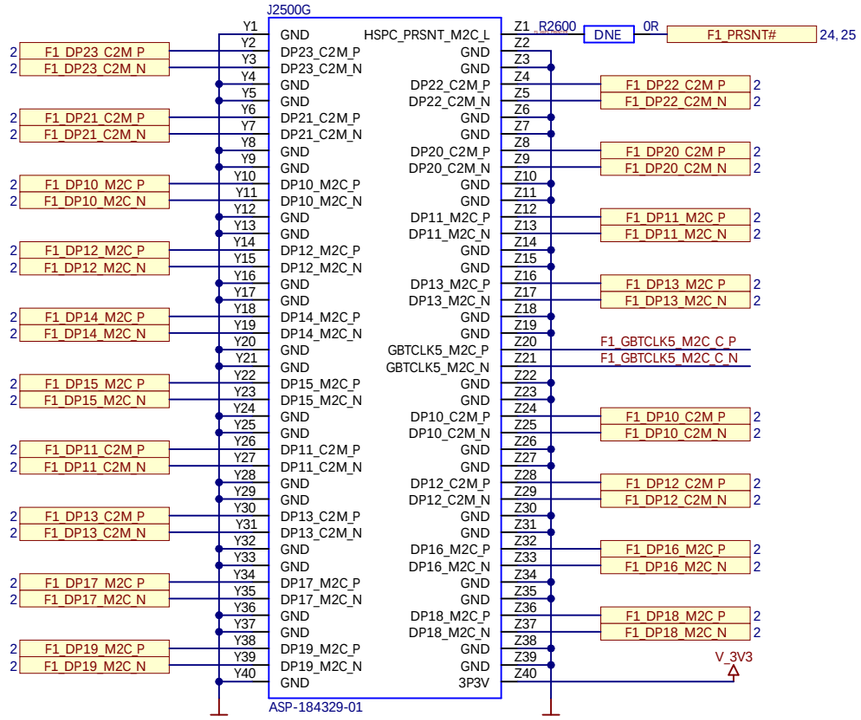




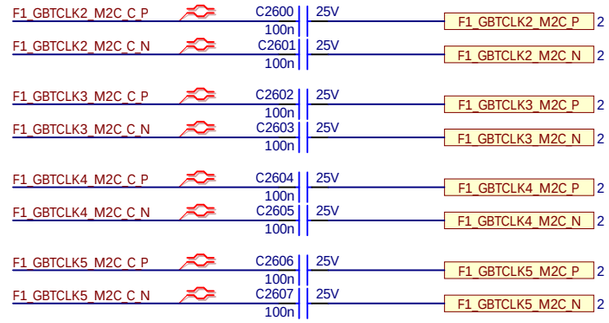
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P25_F1_1.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 25 / 36

FMC HSPC Connector 1 M, L, Y, Z

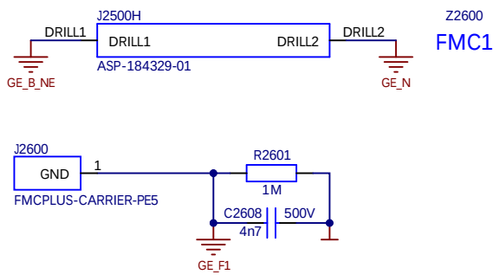




FMC 1 GBTCLK AC Coupling

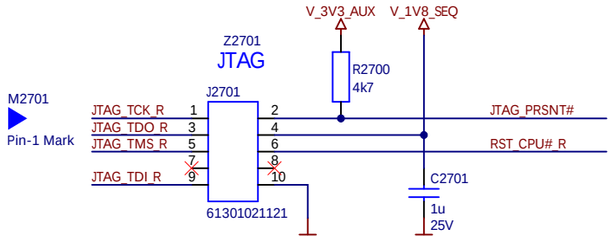


FMC HSPC Connector 1 Mechanics

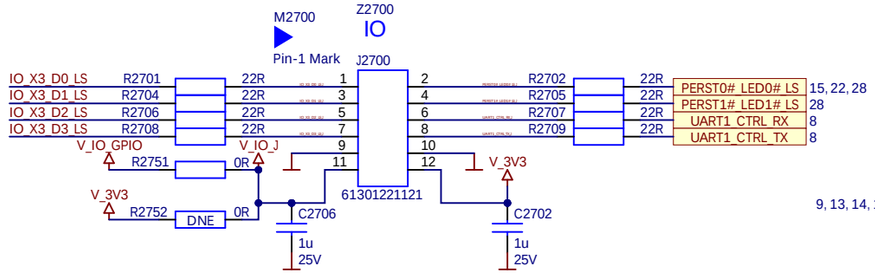


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P26_F1_2.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 26 / 36

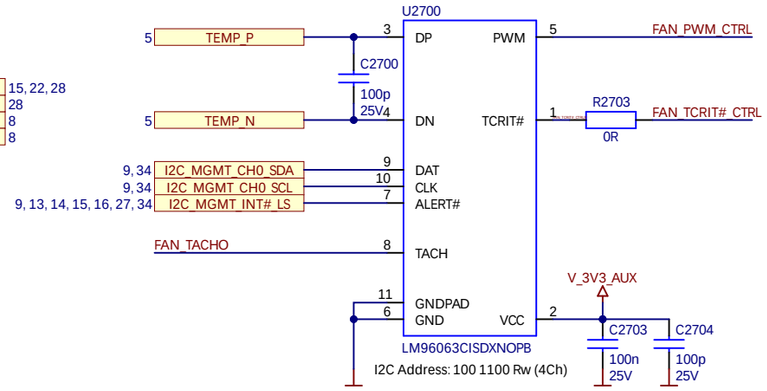
JTAG Connector



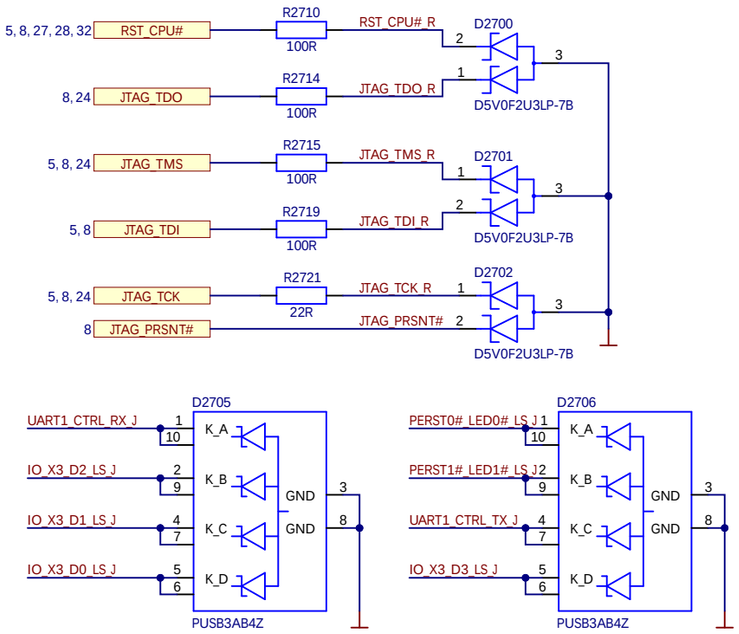
IO Connector



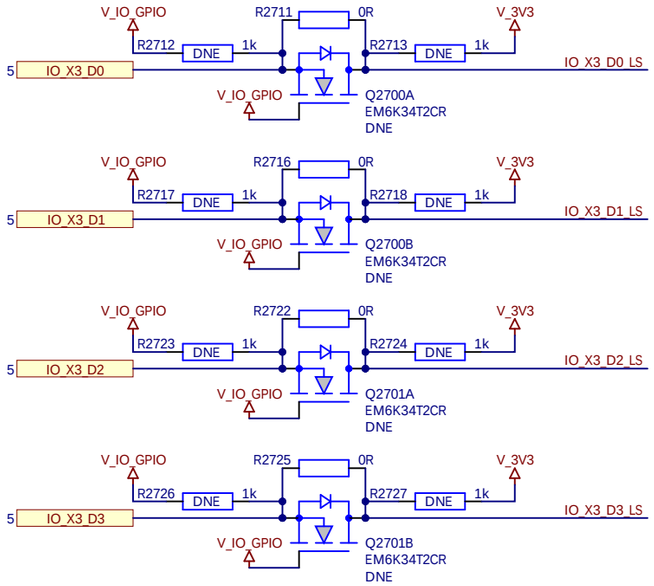
Fan Controller



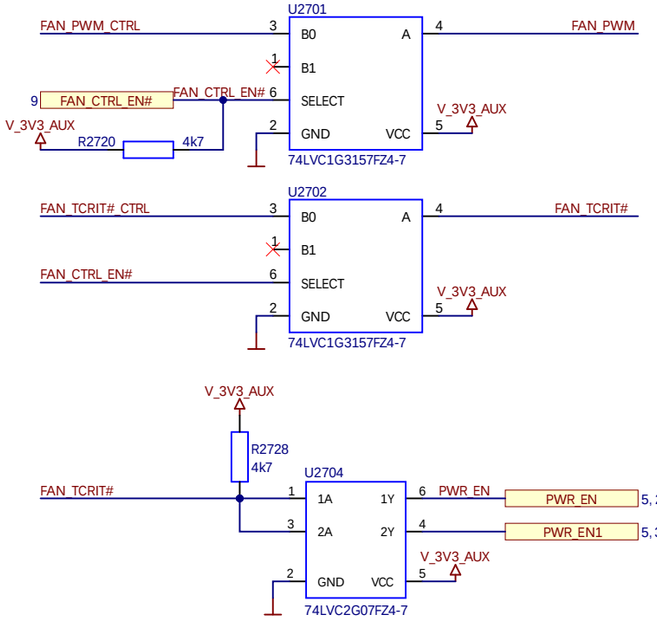
ESD Protection for JTAG and IO Connectors



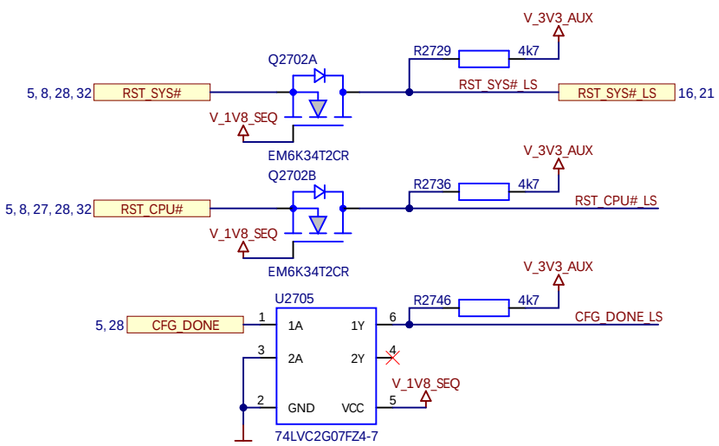
IO Level Shifters



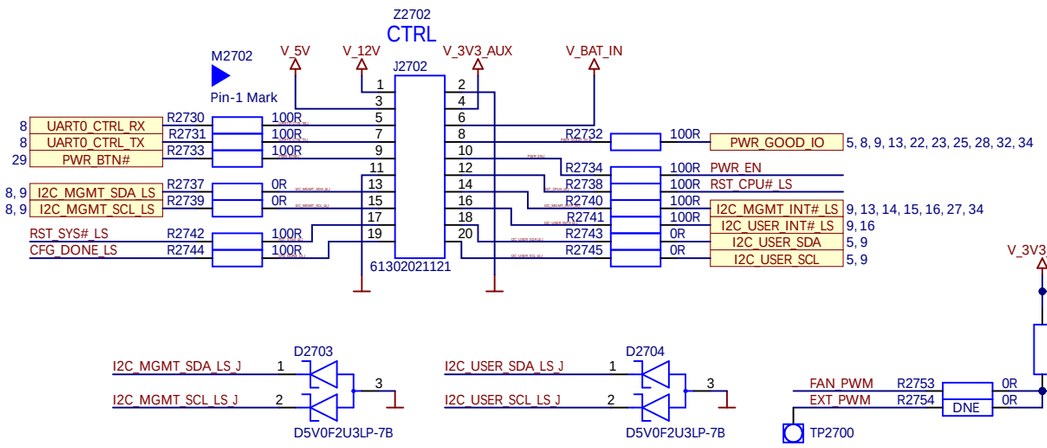
Fan Connector



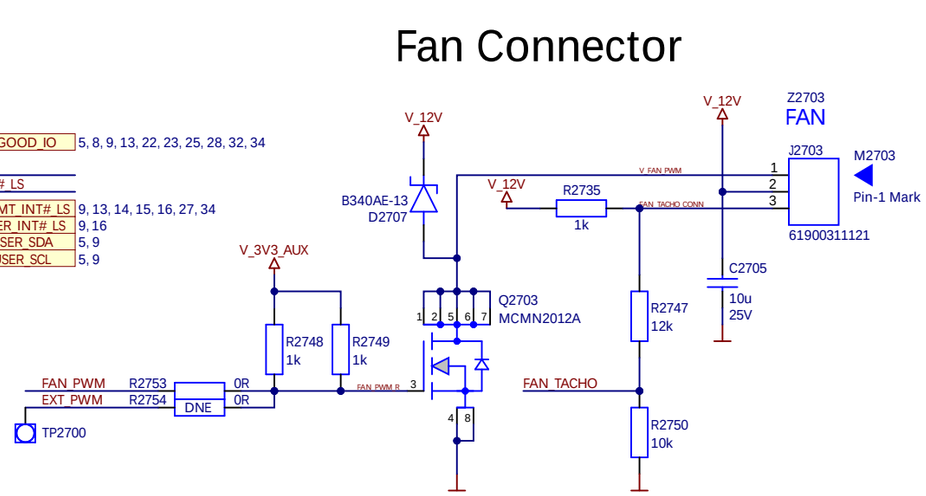
Control Connector Level Shifters



Control Connector

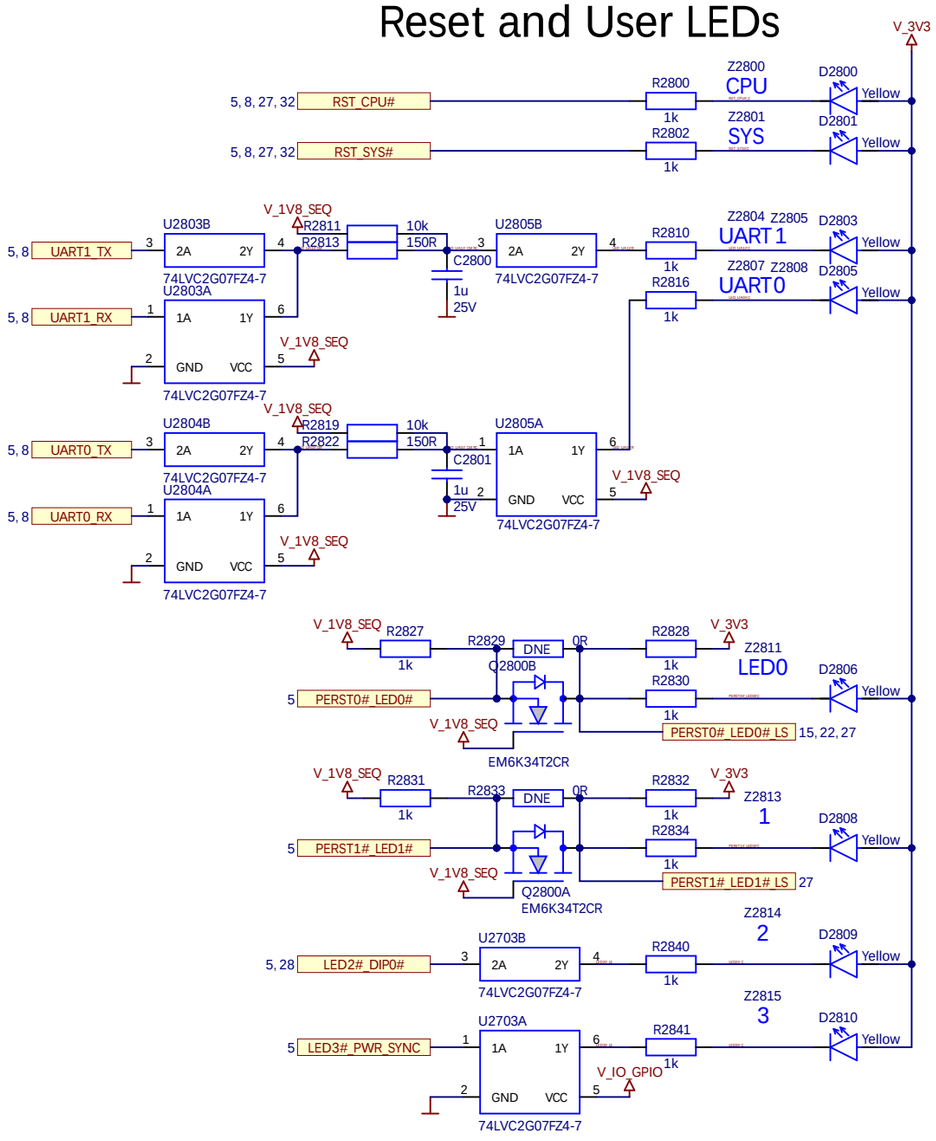


Fan Connector

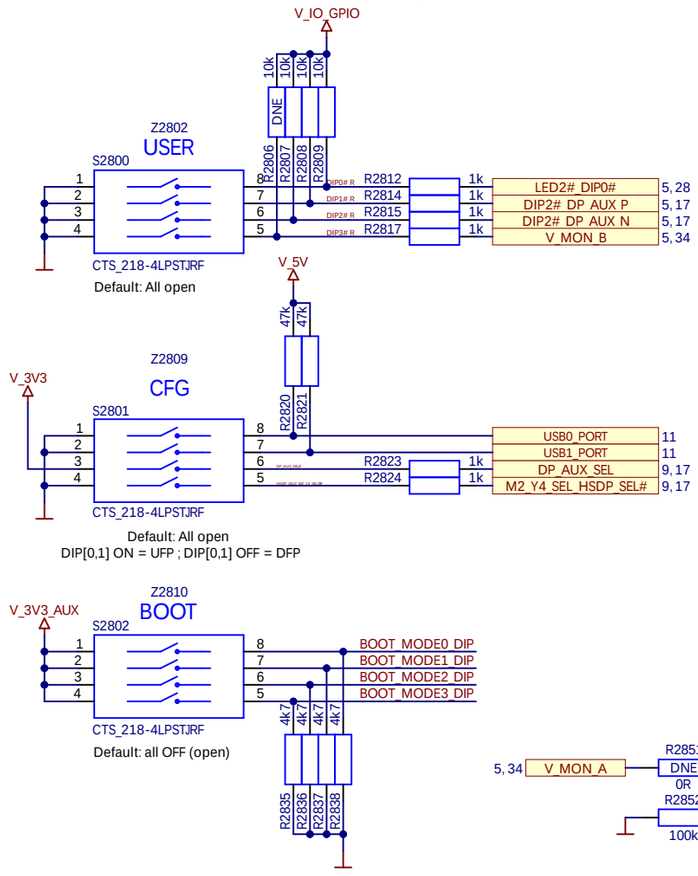


ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P27_JTAG_PMOD_CTRL_FAN.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 27 / 36

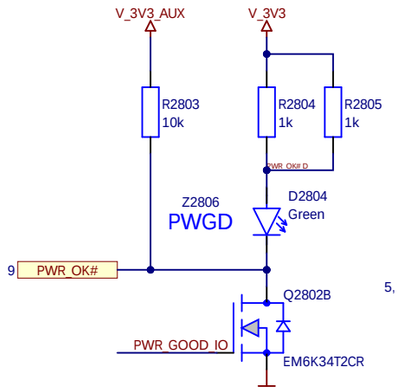
Reset and User LEDs



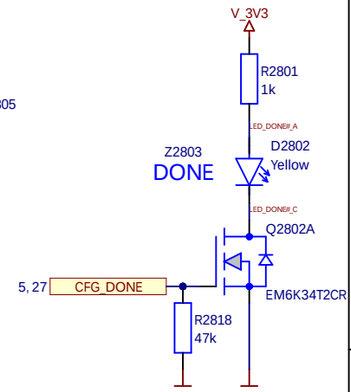
Board Configuration



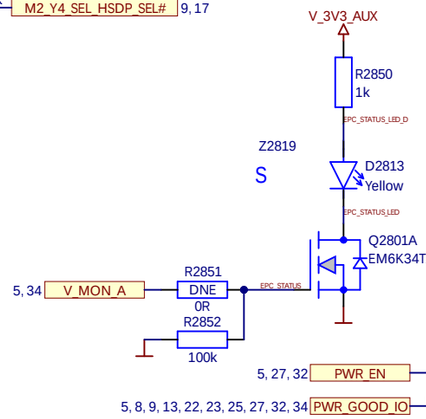
Power Good LED



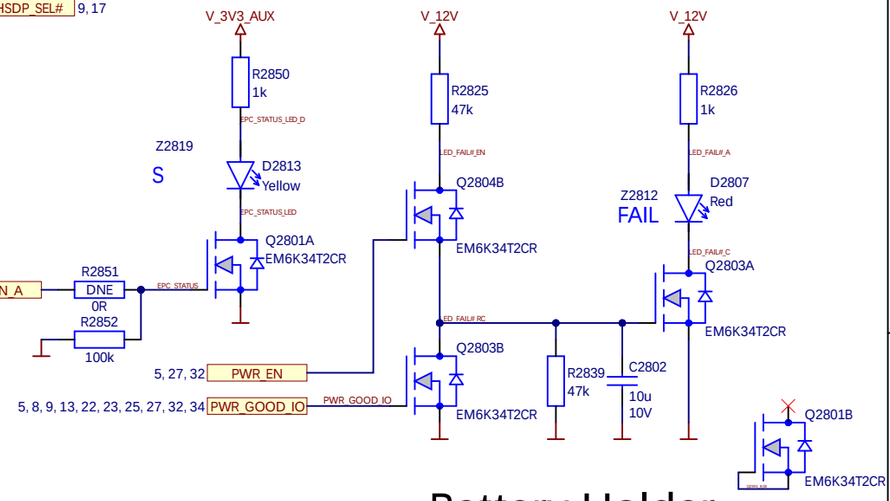
Done LED



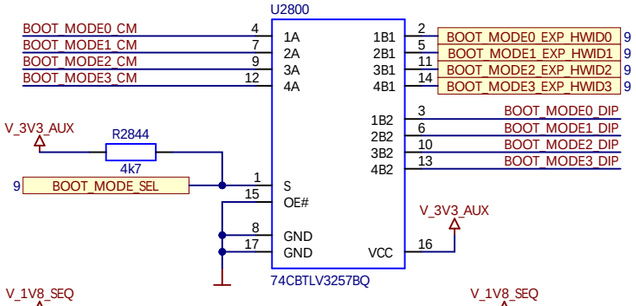
EPC STATUS LED



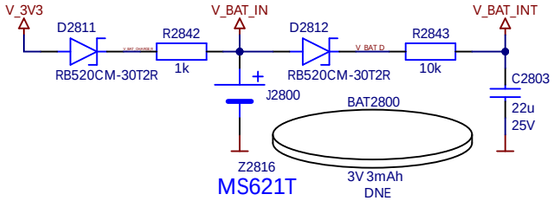
Power Fail LED



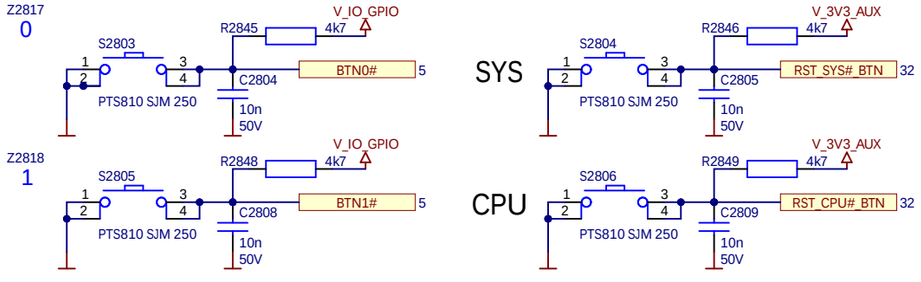
Boot Mode Selection



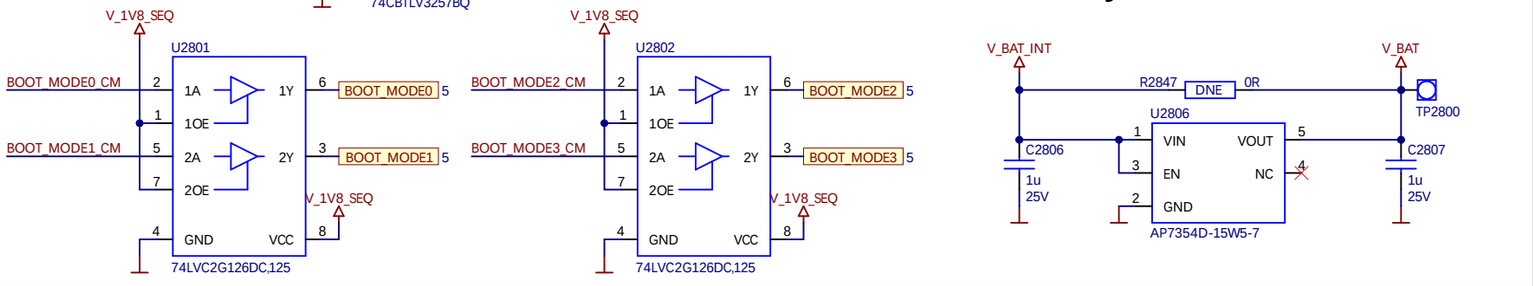
Battery Holder



Buttons



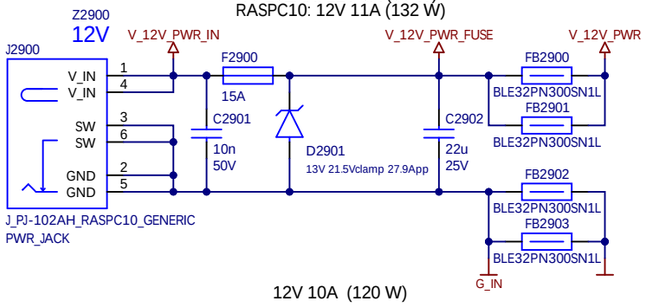
Battery LDO 1.5V 10mA



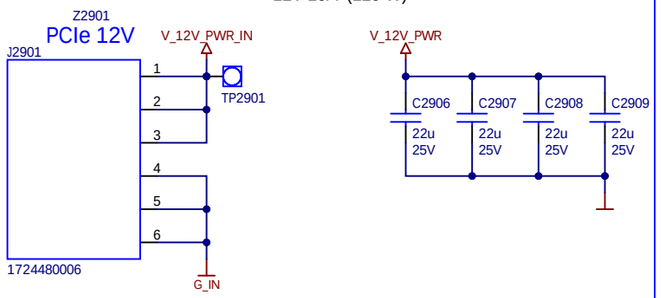
ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P28_User_Interface.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 28 / 36

Power Input Circuit

PJ-102AH: 12V 5A (60 W)
RASPC10: 12V 11A (132 W)



12V 10A (120 W)



Press ON Press OFF Circuit

not included in user schematic

Input Blocking Controller

not included in user schematic

V_12V_PWR I limit = 13.5A

V_12V_PCIE I limit = 6.75A

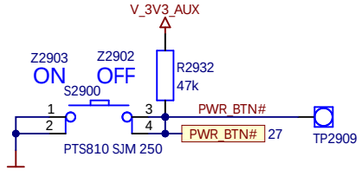
Buck Converter Auxiliary

3.3V 3A 5%
Fs = 500kHz Vfb = 0.8V
not included in user schematic

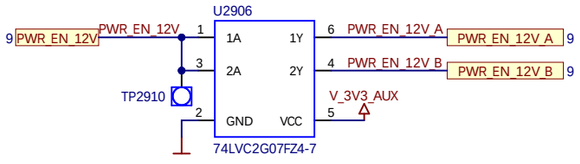
Board Decoupling

not included in user schematic

ON/OFF Button



Reset Signalling



ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P29_Power_Input.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 29 / 36

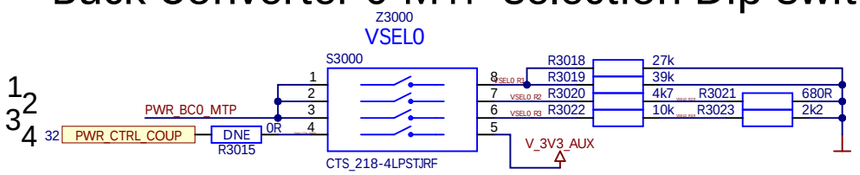
Buck Converter 0

I2C-Address = 1Eh, PMBUS-Address = 4Eh
not included in user schematic

Buck Converter 0 Power Stage

not included in user schematic

Buck Converter 0 MTP Selection Dip Switch



Default: 1110 1.8V

V_IO__F0_OVP	V_ADJ_F0	VSEL0
Off		On 1 2 3 4
1.2V	1.2V	On 1 2 3 4
1.5V	1.5V	On 1 2 3 4
1.57V	1.72V	On 1 2 3 4

V_IO__F0_OVP	V_ADJ_F0	VSEL0
1.8V	1.8V	On 1 2 3 4
2.5V	2.5V	On 1 2 3 4
3.3V	3.3V	On 1 2 3 4
not valid		On 1 2 3 4

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P30_Power_Converter.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 30 / 36

Buck Converter 1

I2C-Address = 17h, PMBUS-Address = 47h
not included in user schematic

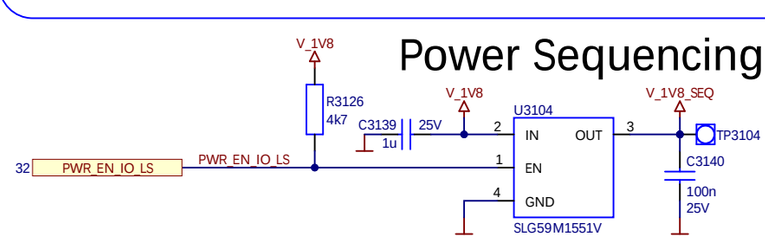
Overtemperature Reset BC0

not included in user schematic

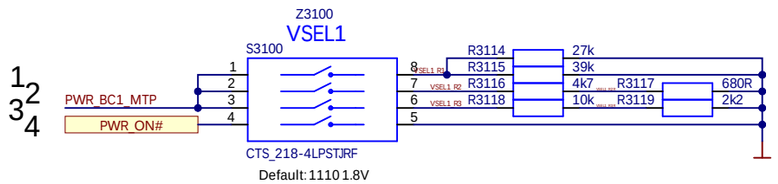
Power Converter Synchronization

not included in user schematic

Power Sequencing



Buck Converter 1 MTP Selection DIP Switch



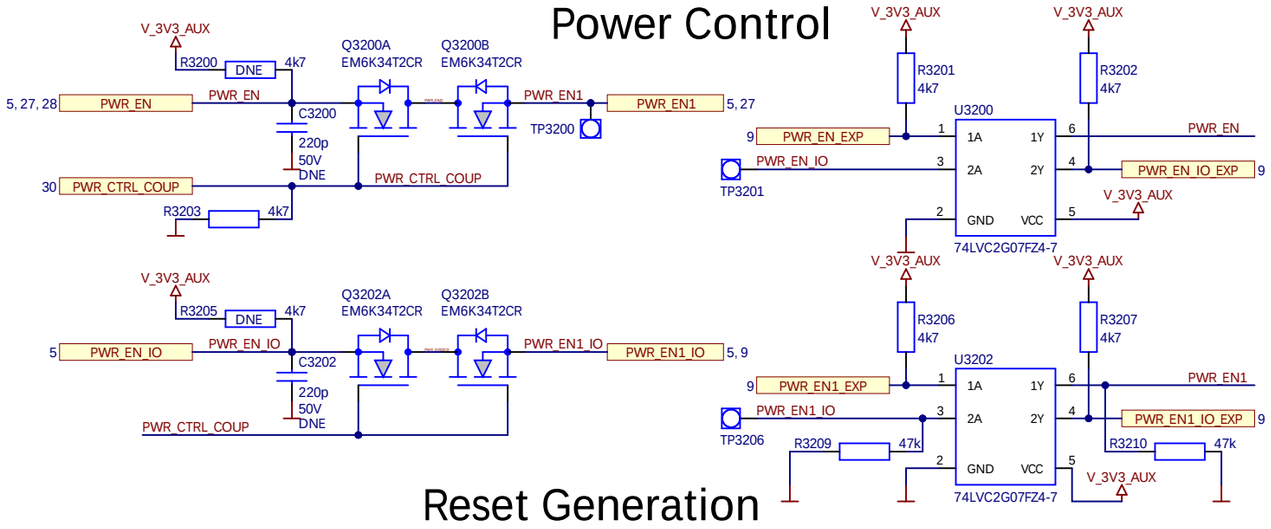
Default: 1110 1.8V

V_IO__F1_OVP	V_ADJ_F1	VSEL1
Off		On 1 2 3 4
1.2V	1.2V	On 1 2 3 4
1.5V	1.5V	On 1 2 3 4
1.57V	1.72V	On 1 2 3 4

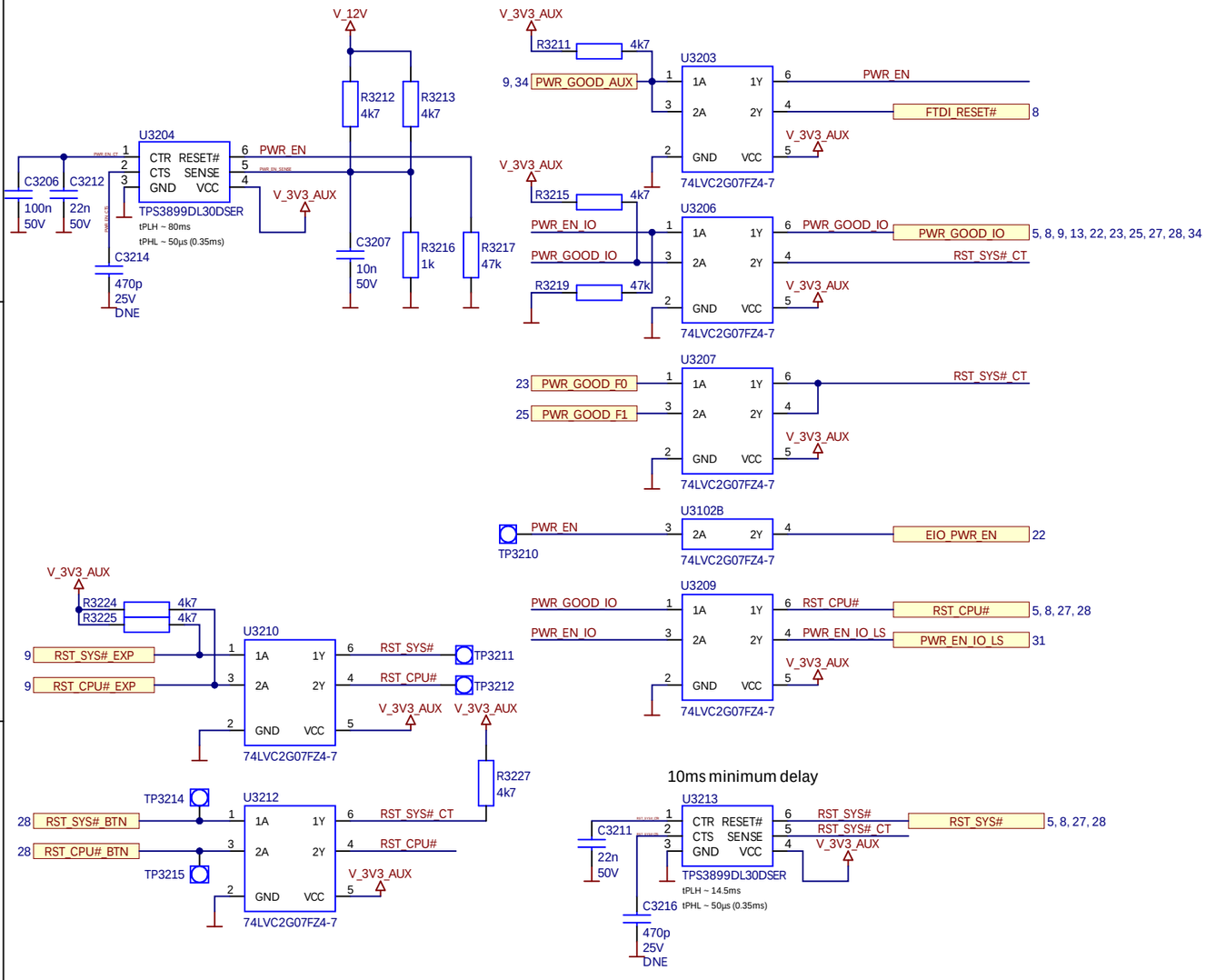
V_IO__F1_OVP	V_ADJ_F1	VSEL1
1.8V	1.8V	On 1 2 3 4
2.5V	2.5V	On 1 2 3 4
3.3V	3.3V	On 1 2 3 4
not valid		On 1 2 3 4

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P31_Power_Converter.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 31 / 36

Power Control



Reset Generation



IO Over-Voltage Protection

not included in user schematic

Power Cycle Circuit

not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P32_Reset_OVP.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 32 / 36

6 5 4 3 2 1

D C B A

Module Power Load Switches

not included in user schematic

3.3V MOD1 Load Switch

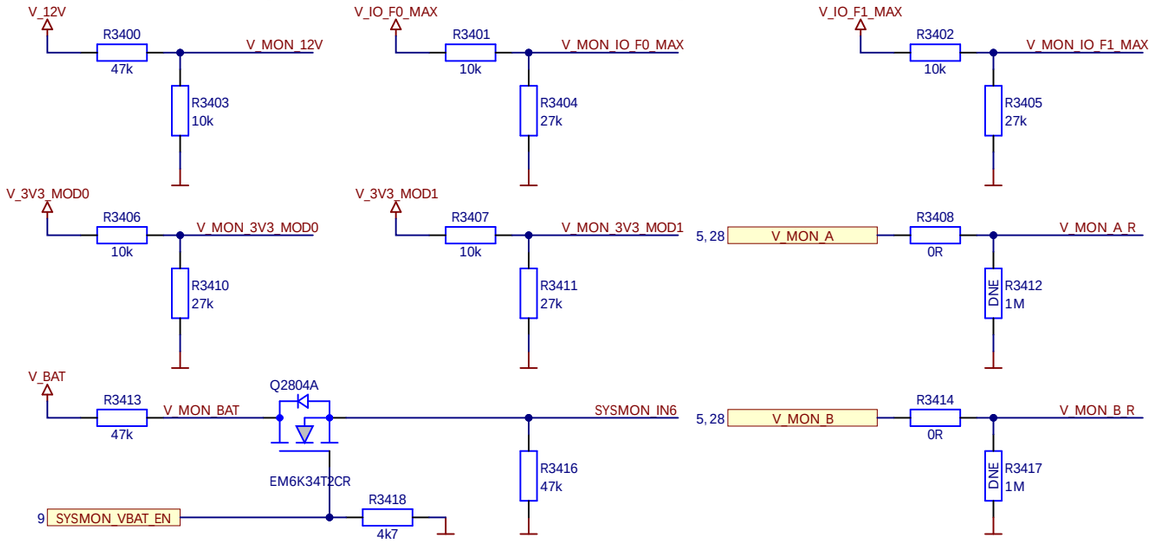
not included in user schematic

3.3V MOD0 Load Switch

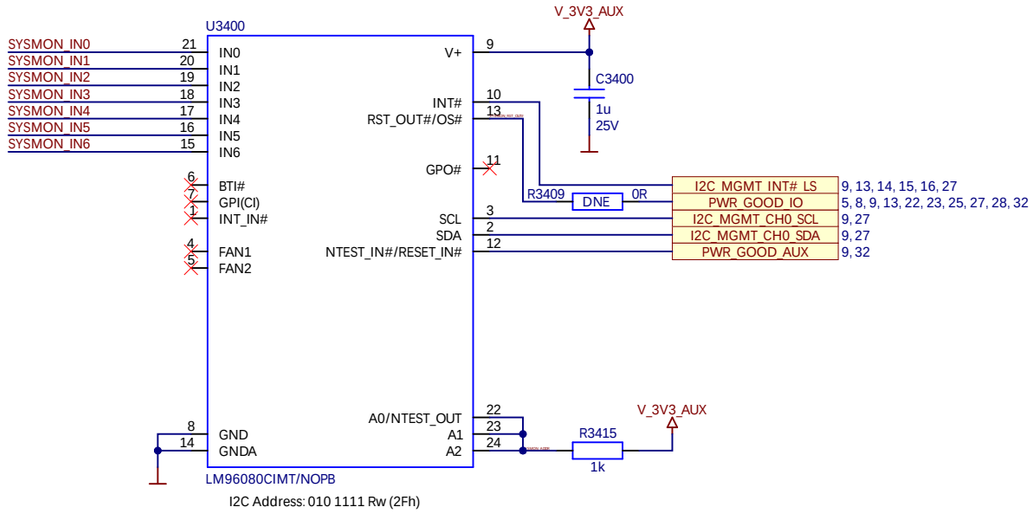
not included in user schematic

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P33_Power_Module.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 33 / 36

System Monitor Voltage Sense



System Monitor

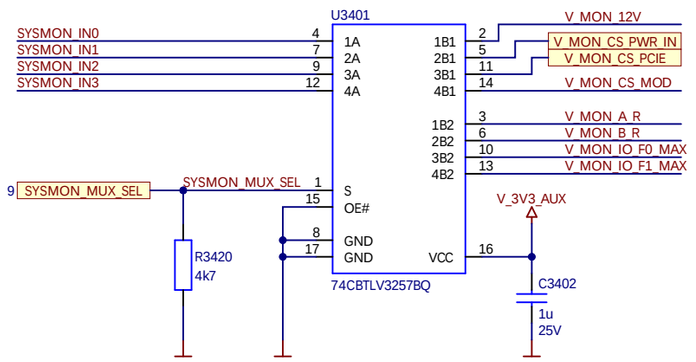


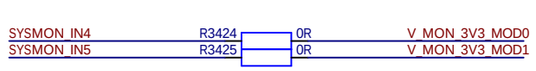
Analog Current Sensors

not included in user schematic

Gain 100V/V V_MON_CS_MOD=I_MOD*0.1V/A

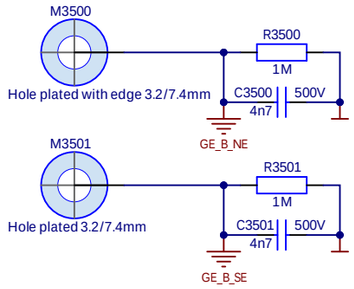
System Monitor Analog Multiplexers

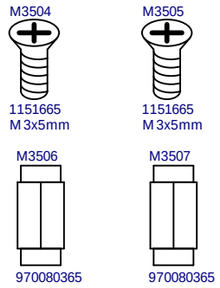




ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P34_System_Monitor.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 34 / 36

Board Mounting Holes

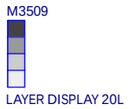




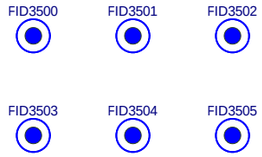
Texts, Logos and Labels



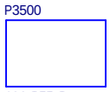
Layer Display



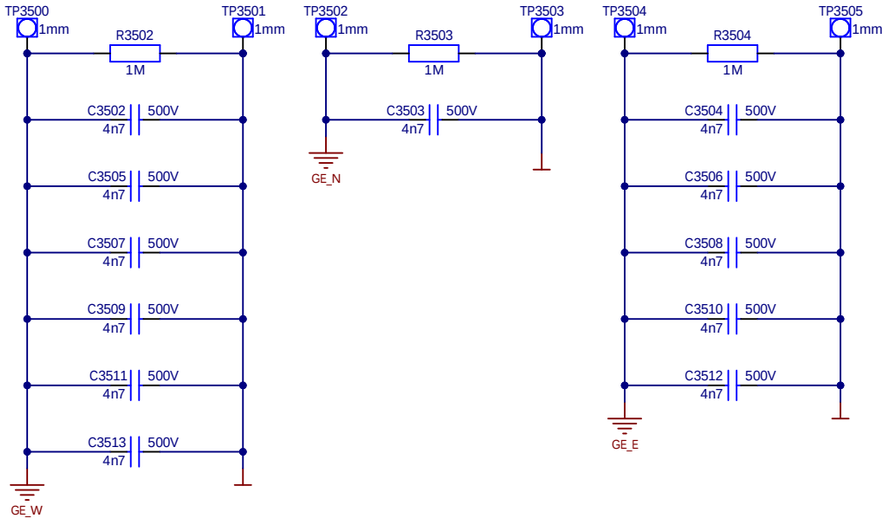
PCB Fiducials



PCB



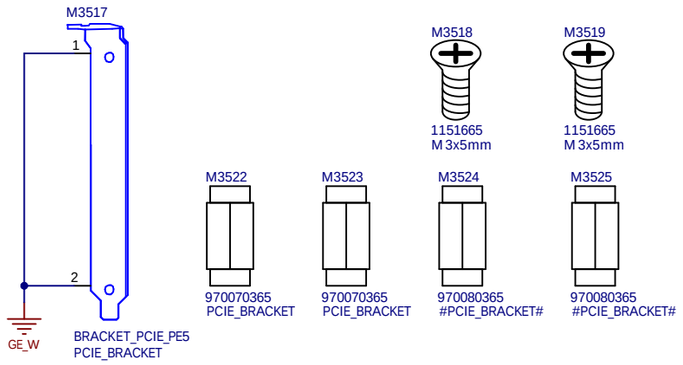
Chassis Ground



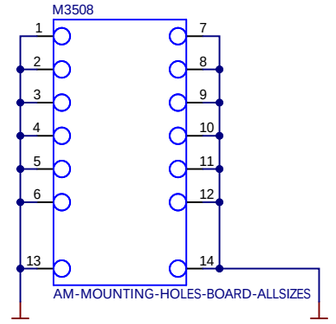
Ground Test Points

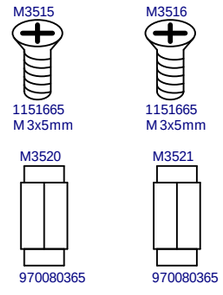


PCIe Bracket



Andromeda Module Mounting Holes





ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P35_Mechanics.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 35 / 36

Component Options

PWR_JACK	L3U3_3A	L3U3 (2A)	L2U2 (4A)	SFP	PCIE_BRACKET		
Part J2900A PJ-102AH EN100388 Part J2900B RASPC10 EN104008	Part L2900A 3u3 EN103135	Part L3001A 3u3 EN103135 Part L3001B 3u3 EN103134 Part L3001C 3u3 EN103132 Part L3101A 3u3 EN103135 Part L3101B 3u3 EN103134 Part L3101C 3u3 EN103132 Part L3100A 3u3 EN103135 Part L3100B 3u3 EN103134 Part L3100C 3u3 EN103132	Part L3002A 2u2 EN102185 Part L3002B 2u2 EN103281 Part L3102A 2u2 EN102185 Part L3102B 2u2 EN103281 Part L3003A 2u2 EN102185 Part L3003B 2u2 EN103281 Part L3103A 2u2 EN102185 Part L3103B 2u2 EN103281	Part J1600A 2170088-2 EN104118 Part J1601A 2170088-2 EN104118	M3517, M3522, M3523 or M3524, M3525		

Component Assembly

Enclustra Product	PWR_JACK	L3U3_3A	L3U3 (2A)	L2U2 (4A)	SFP	PCIE_BRACKET		
Variant EN106914:0 AM-PE5-W-R1.1	Option EN106914:1 PWR_JACK EN100388	Option EN106914:2 L3U3_3A EN103135	Option EN106914:3 L3U3 EN103135	Option EN106914:4 L2U2 EN102185	Option EN106914:5 SFP EN104118	Option EN106914:6 PCIE_BRACKET DNE		

ENCLUSTRA FPGA SOLUTIONS	Project AM-PE5.PrjPCB	Sheet Name AM-PE5_P36_Variants.SchDoc	Customer No 0000	Revision R1.1	DNE = Do Not Equip	Confidential
	Company Enclustra FPGA Solutions	Copyright © 2025 by Enclustra FPGA Solutions	Project No 473	Designed BOEL	Date 17 Apr 2025	Sheet/sheets 36 / 36